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MSc Data Analytics

Predictive Performance Versus Interpretability in Machine Learning Models for Telecom Customer Churn: A Comparative Evaluation Using SHAP Analysis

By

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*A dissertation submitted in partial fulfilment of the requirements for the degree of Master of
Science in Data Analytics*

29th May, 2026

DECLARATION

I hereby declare that the work contained in this dissertation is my own and has not been submitted for any other academic award. Specifically, I confirm the following:

- 1. Originality:** This dissertation is submitted in partial fulfilment of the requirements for the degree of Master of Science in Data Analytics at De Montfort University, Leicester. The research presented is original work undertaken by me during the period of my registration on the programme.
- 2. Non-duplication:** No part of this dissertation has previously been submitted to this or any other institution for a degree, diploma, or any other qualification.
- 3. Academic integrity:** All sources consulted in the preparation of this dissertation have been duly cited and acknowledged in accordance with the Harvard referencing system. Any material reproduced from published or unpublished sources has been clearly indicated.
- 4. Data and code:** The dataset used in this study, the Iranian Telecom Churn dataset, is publicly available via the UCI Machine Learning Repository and is acknowledged accordingly. The analytical pipeline and visualisation code were written by me and are reproducible from the submitted Jupyter notebook.

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ABSTRACT

Customer churn in telecommunications presents a major profitability challenge in which retention decisions increasingly rely on predictive machine learning systems. Ensemble methods deliver substantial discrimination gains, but their opacity raises legitimate concerns for operational and regulatory contexts. This dissertation evaluates logistic regression, Random Forest, and XGBoost on telecom customer churn data, assessing predictive performance and interpretability jointly through a structured SHAP analysis.

Using the Iranian Telecom Churn dataset ($n = 3,150$; 15.65% churn rate), a methodologically rigorous pipeline was developed incorporating leakage-free stratified cross-validation, in-fold SMOTE resampling, randomised hyperparameter search, and calibration diagnostics. Performance was evaluated using discrimination metrics (PR-AUC, AUC-ROC, F1, MCC) appropriate for imbalanced classification, alongside Brier scores for probability reliability. SHAP values were computed using TreeSHAP for the best-performing ensemble model and benchmarked directly against logistic regression coefficients via a comparative ranking framework, with bootstrap stability analysis across 100 resamples.

XGBoost achieved the strongest performance across all primary metrics (PR-AUC = 0.922, F1 = 0.812, Brier = 0.050), with the performance hierarchy (LR < RF < XGBoost) confirmed by repeated cross-validation and robust to three alternative feature regimes. The SHAP-versus-coefficient comparison identified five features in strong agreement, four in moderate agreement, and two in substantive divergence: Subscription Length exhibited a U-shaped non-linear effect invisible to the linear baseline, consistent with Switching Cost Theory, while Complains showed a strong conditional effect diluted at the population level by its low base rate. A cost-sensitive threshold optimisation combined with Customer Lifetime Value

segmentation produced a deployment framework yielding 2.20 times the expected retained value of a highest-risk-first targeting strategy under equivalent intervention budgets.

The principal contribution is the repositioning of SHAP from a descriptive output of opaque models to a diagnostic instrument for identifying where linear-model assumptions fail. Findings have direct implications for the deployment of explainable retention decision systems in telecommunications and beyond.

Keywords: Customer churn; telecommunications; logistic regression; Random Forest; XGBoost; SHAP; explainable AI; calibration; Customer Lifetime Value; cost-sensitive learning.

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LIST OF ABBREVIATIONS AND TERMINOLOGIES

AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CLV	Customer Lifetime Value
CV	Cross-Validation; Coefficient of Variation (context-dependent)
EDT	Expectation-Disconfirmation Theory
F1	F1 Score (harmonic mean of precision and recall)
FN	False Negative
FP	False Positive
HL	Hosmer–Lemeshow (calibration test)
IBM Telco	IBM Telecommunications Customer Churn dataset
LIME	Local Interpretable Model-Agnostic Explanations
LR	Logistic Regression
MCC	Matthews Correlation Coefficient
MSc	Master of Science
OR	Odds Ratio
PR-AUC	Precision–Recall Area Under the Curve
RF	Random Forest
ROC	Receiver Operating Characteristic
RQ	Research Question
SCT	Switching Cost Theory
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
TN	True Negative
TP	True Positive
UCI	University of California, Irvine (Machine Learning Repository)
VIF	Variance Inflation Factor
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Customer churn, the voluntary discontinuation of a service by a subscriber, is one of the most economically significant challenges facing telecommunications operators. In a saturated, price-competitive industry, retention is consistently more cost-effective than acquisition (Reichheld and Schefer, 2000); lost customer lifetime value compounds with forfeited future revenue and replacement acquisition costs (Verbeke et al., 2012). Annual churn rates of 15% or higher are regularly reported in the global mobile sector (Burez and Van den Poel, 2009).

One element of the solution is predictive analytics. With the use of machine learning models, organisations can predict the probability of subscriber churning and act accordingly. This study draws on three theoretical frameworks. Oliver's (1980) Expectation–Disconfirmation Theory states that when a customer's expectations are not met, they will defect. Per Porter's Switching Cost Theory (1980), the ease of switching is moderated by structural variables such as contractual lock-in, portability friction, and service dependency. Furthermore, Customer Lifetime Value theory (Gupta et al., 2004) adds a commercial perspective that not all losses are equally costly and that a good churn system combines probability, risk segment and business value into a single choice.

Machine learning approaches to churn prediction have progressed through three generations. Logistic regression was used in early work for its statistical interpretability since each predictor is expressed in terms of odds ratios, which capture the direction and magnitude of each predictor in a clear inferential framework (Lemmens and Croux, 2006). Interpretability is not merely academic; regulators increasingly require automated decisions to be explainable. Second-generation methods, using ensembles of models without requiring linearity assumptions, such as Random Forest (Breiman, 2001) and gradient boosting, such as

XGBoost (Chen and Guestrin, 2016), aimed to improve discriminative performance at the cost of interpretability. The third generation recovers interpretability through post-hoc explanation, most notably SHAP (Lundberg and Lee, 2017), which grounds feature attribution in cooperative game theory and provides consistent decompositions at global and individual levels. This evolution raises a central question: whether the predictive advantages of ensembles can be reconciled with the interpretability required for operational decision-making.

1.2 Problem Statement

Despite the growing application of SHAP to churn prediction, the existing literature reveals a consistent limitation in how interpretability is treated. Prior studies tend to apply SHAP descriptively to a single model in isolation, using it to report which features matter to the ensemble rather than as a diagnostic instrument for understanding what the ensemble has learned that a simpler model cannot represent. This produces two mutually incomplete literatures: an interpretable-but-limited logistic regression literature that cannot capture nonlinear churn dynamics, and a performant-but-opaque ensemble literature that cannot explain the nature or source of its performance advantage. The analytical value of bridging these two literatures (by comparing SHAP attributions directly against logistic regression coefficients) has received limited attention in published work.

A further limitation is methodological. Studies on the Iranian Telecom Churn dataset (Jafari-Marandi et al., 2020) have not consistently addressed the conditions under which results can be trusted: predicted probability calibration, stability of feature importance rankings across resamples, and the data leakage risks introduced by applying resampling procedures before rather than within cross-validation folds (Blagus and Lusa, 2013). In the absence of these diagnostics, reported performance metrics can be the result of pipeline aberrations rather than

a true predictive signal. Lastly, a lot of published research optimises for a single performance metric without converting model outputs into the decision-relevant quantities that a retention team actually uses, such as customer value segmentation, cost-sensitive prioritisation, and classification thresholds (Neslin et al., 2006; Devriendt et al., 2021).

1.3 Research Aim

This study conducts a comparative evaluation of logistic regression, Random Forest, and XGBoost for telecom customer churn prediction, assessing predictive performance and interpretability through structured SHAP analysis and translating findings into business decision tools grounded in Customer Lifetime Value theory. The central thesis is that post-hoc explanation methods can function as analytical instruments for diagnosing where linear model assumptions fail, rather than merely describing the predictions of opaque models.

1.4 Research Objectives

The study is guided by four objectives:

1. To construct a reproducible and leakage-free churn prediction pipeline for logistic regression, Random Forest, and XGBoost, incorporating preprocessing, class imbalance handling, and robustness validation across cross-validation, resampling, and feature regimes.
2. To compare logistic regression, Random Forest, and XGBoost in terms of predictive performance using discrimination metrics and reliability-based calibration diagnostics appropriate for imbalanced classification problems.
3. To evaluate model interpretability by comparing logistic regression coefficient interpretations with SHAP-derived feature attributions from the best-performing ensemble model.

4. To develop a cost-sensitive retention decision framework that integrates churn probability estimates with customer lifetime value.

1.5 Research Questions

Three research questions underpin the objectives set out in Section 1.4. Where the objectives specify the operational steps of the investigation, the questions identify the conceptual unknowns those steps are designed to resolve.

- RQ1. To what extent do ensemble methods outperform logistic regression in telecom churn prediction when evaluated using discrimination and calibration metrics appropriate for imbalanced classification problems?
- RQ2. Where do SHAP-derived feature attributions from the best-performing ensemble model diverge from logistic regression coefficients, and how are those divergences explained through the behavioural frameworks of Expectation-Disconfirmation Theory and Switching Cost Theory?
- RQ3. How can churn prediction outputs be translated into operational retention decisions that incorporate customer lifetime value and the asymmetric costs of misclassification?

1.6 Scope of the Study

The study is bounded in four respects. First, by data: the Iranian Telecom Churn dataset (UCI ML Repository #563) contains 3,150 data points (2,850 after de-duplication) with 15 behavioural and demographic attributes and a binary churn label aggregated over a 12-month window. The cross-sectional design provides a single snapshot from a single Iranian provider in approximately 2010, and findings are not necessarily generalisable to contemporary Western markets. This constraint is somewhat mitigated by the fact that the underlying

behavioural mechanisms (service dissatisfaction, switching costs, value asymmetry) are theoretical, not market-specific, an assumption tested in Chapter 5 through cross-study comparison.

Second, by methodology: the study compares logistic regression, Random Forest, and XGBoost with SHAP as the post-hoc explanation method. LIME, counterfactual explanations, attention-based methods, neural networks, LightGBM, and CatBoost are out of scope. The choice of gradient-boosted ensembles over deep architectures reflects empirical evidence that tree-based methods remain state-of-the-art on medium-sized tabular data of the order of 10,000 samples (Grinsztajn et al., 2022). Third, by analytical objective: the study addresses static churn classification at a single decision point, not sequential or survival modelling. Fourth, by deployment: cost matrix parameters are derived from the dataset's Customer Value distribution rather than empirical business data, and no live A/B testing or causal intervention evaluation is undertaken.

1.7 Dissertation Structure

Chapter 2 reviews the literature on customer churn prediction, machine learning interpretability and SHAP analysis, identifying the methodological and theoretical gaps motivating this research. Chapter 3 describes the methodology. It includes data preparation, the modelling pipeline and the evaluation framework. Chapter 4 reports results across four stages of analysis (logistic regression baseline, ensemble model comparison, SHAP interpretability, and decision optimisation), plus calibration, robustness and deployment simulation. Chapter 5 discusses the findings with reference to the research questions, the theoretical framework and previous literature including cross-market generalisability. Chapter 6 reviews contributions, quantifies limitations, and proposes directions for future research.

CHAPTER 2: THEORETICAL FRAMEWORK AND LITERATURE REVIEW

This chapter synthesises the academic literature across six areas underpinning this dissertation. The review proceeds from industry context and theoretical explanation of churn behaviour, through predictive modelling approaches and model evaluation methodology, to interpretability frameworks and the synthesis of identified research gaps.

2.1 Customer Churn in the Telecommunications Industry

The telecommunications sector represents one of the most commercially significant contexts for churn prediction research. Long-term profitability in markets with high penetration and commoditised pricing is driven by retention rather than acquisition. The compounding loss of customer lifetime value means modest reductions in churn rate produce disproportionate revenue gains (Reichheld and Schefter, 2000). Published figures vary depending on the maturity of the market, the regulatory climate and the level of competition, but rates of 15% or more are regularly recorded (Burez and Van den Poel, 2009; Verbeke et al., 2012).

A common finding in the literature is that churn risk is concentrated among consumers who experience service quality failures and persistently low engagement (Verbeke et al., 2012). These behavioural signals are observable in operator data prior to the churn event, which is the foundational premise of predictive modelling. Neslin et al. (2006) highlighted a gap between statistical modelling practice and operational deployment, arguing that discriminative performance alone is insufficient for commercial utility, a concern that motivates this study's integration of cost-sensitive threshold optimisation and CLV segmentation. Expanding on this idea, Devriendt et al. (2021) argued that churn prediction can be reframed as an issue of optimising business value instead of a classification challenge, which would lead to significantly better practical returns from retention programs.

2.2 Theoretical Foundations of Customer Churn

Predictive models detect statistical correlations between observable features and churn outcomes but do not explain why such correlations exist. Without theoretical grounding, a model cannot distinguish a true causal mechanism from a spurious association or guide interventions that address underlying causes. The three frameworks below combine to provide a comprehensive account of why telecom customers churn.

2.2.1 Expectation-Disconfirmation Theory

According to Oliver's (1980) Expectation-Disconfirmation Theory (EDT), satisfaction is not a function of absolute service quality but of the difference between expected and actual performance. Negative disconfirmation raises defection likelihood, while positive disconfirmation promotes retention. In telecoms EDT discusses how network stability, billing correctness and timeliness of support impacts churn (Lemon and Verhoef, 2016). A formal complaint operationalises a threshold disconfirmation event, consistent with complaint behaviour being a strong churn predictor (Umayaparvathi and Iyakutti, 2016; Ullah et al., 2019). EDT also implies that recurring low-level service failures, such as call failure rate, may accumulate into chronic dissatisfaction even without a formal complaint, making both variables theoretically relevant.

2.2.2 Switching Cost Theory

Porter's (1980) Switching Cost Theory explains why discontented customers often do not act. Retention is asymmetrical to satisfaction because of contractual penalties, number portability friction, and cognitive effort to evaluate alternatives. Jones et al. (2000) demonstrated empirically that switching costs moderate the satisfaction-retention relationship, with higher switching costs sustaining retention even when satisfaction is low. A theoretical implication is a nonlinear tenure-churn relationship: risk may be elevated early in the customer lifecycle

before contractual switching costs accrue, lower mid-contract when lock-in is greatest, and re-elevated at contract renewal when the barrier temporarily dissolves. This nonlinear structure cannot be captured by logistic regression's constant linear log-odds slope, motivating both the Box-Tidwell diagnostic and the theoretical interpretation of observed LR–SHAP divergences for Subscription Length.

2.2.3 Customer Lifetime Value Theory

Gupta et al. (2004) formalised CLV theory for retention decision-making, establishing that retention ROI depends not on churn probability alone but on the product of probability and the discounted present value of future revenue. A high-risk, low-value customer is a lower retention priority than a moderate-risk, high-value customer. Elkan (2001) derived the cost-theoretic optimal classification threshold as the ratio of intervention cost to total misclassification cost, showing that when missing a churner substantially exceeds false-positive cost, the optimal threshold lies below the standard 0.50. Bahnsen et al. (2015) demonstrated that cost-sensitive frameworks can yield up to 26.4% improvements in cost savings over cost-insensitive approaches. The cost-sensitive approach implemented herein adheres to existing practices in telecommunications, where cost-sensitive classifiers have shown to be beneficial in minimising false-negative misclassification mistakes that have a disproportionate impact on business (Thakkar et al., 2022).

These three frameworks provide theoretical justification for the behavioural and service-related variables used in churn modelling and establish the interpretive lens through which results are evaluated. Each maps to a domain of observable predictors: service failure indicators (Call Failures, Complaints), contractual and engagement variables (Subscription Length, Tariff Plan, Frequency of Use), and customer value, which motivates Customer Value as a business decision variable (Figure 2.1).

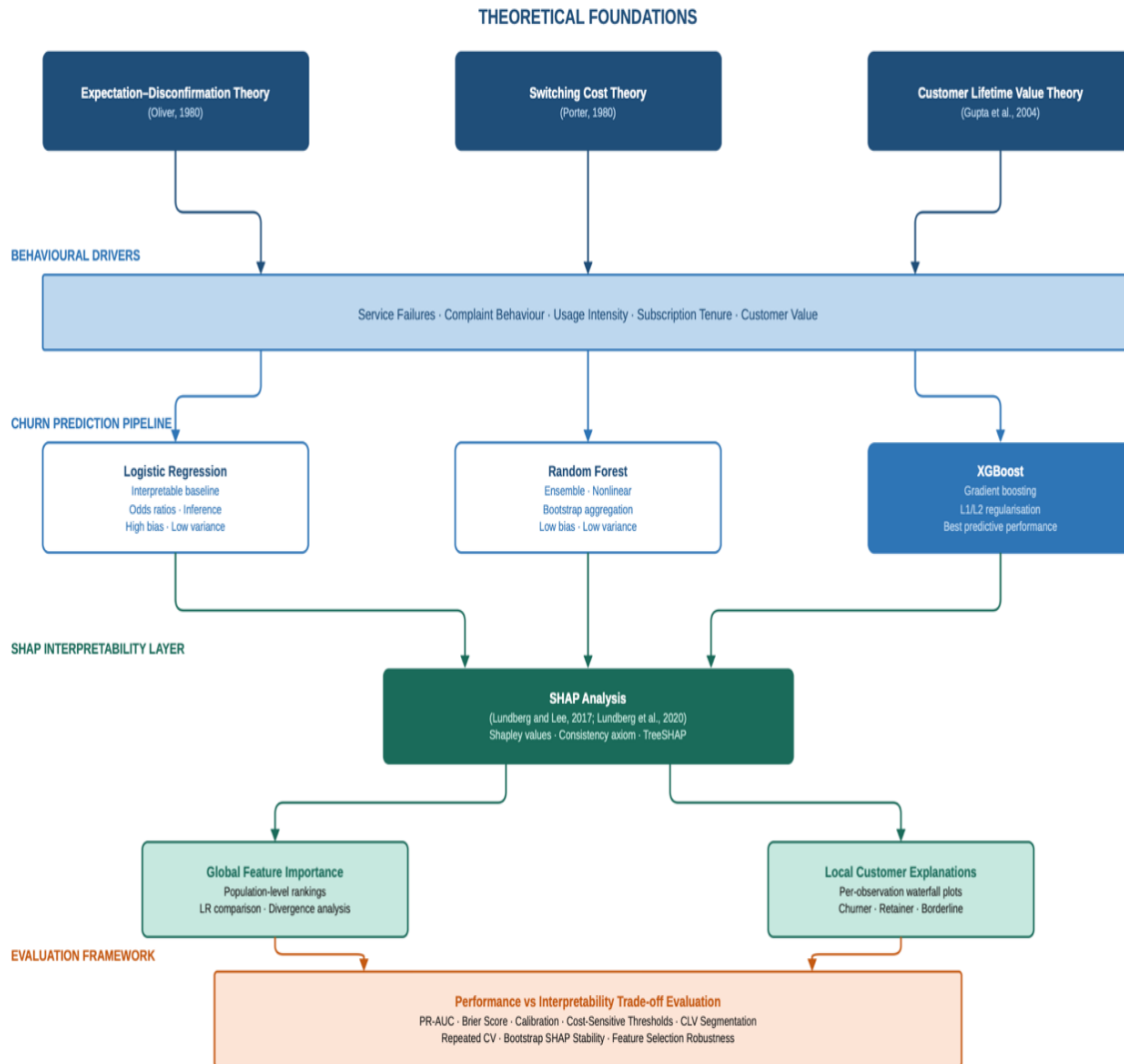


Figure: Conceptual Framework — Theoretical Foundations to Analytical Pipeline

Figure 2.1: Conceptual framework linking theoretical foundations to the analytical pipeline.

2.3 Machine Learning Approaches to Churn Prediction

The literature has evolved through three generations: interpretable linear models; ensembles relaxing linearity constraints for stronger discrimination; and post-hoc methods recovering interpretability from black-box models. Figure 2.2 summarises this taxonomy and the interpretability–performance trade-off that motivates the comparative approach. The three model classes are evaluated below.

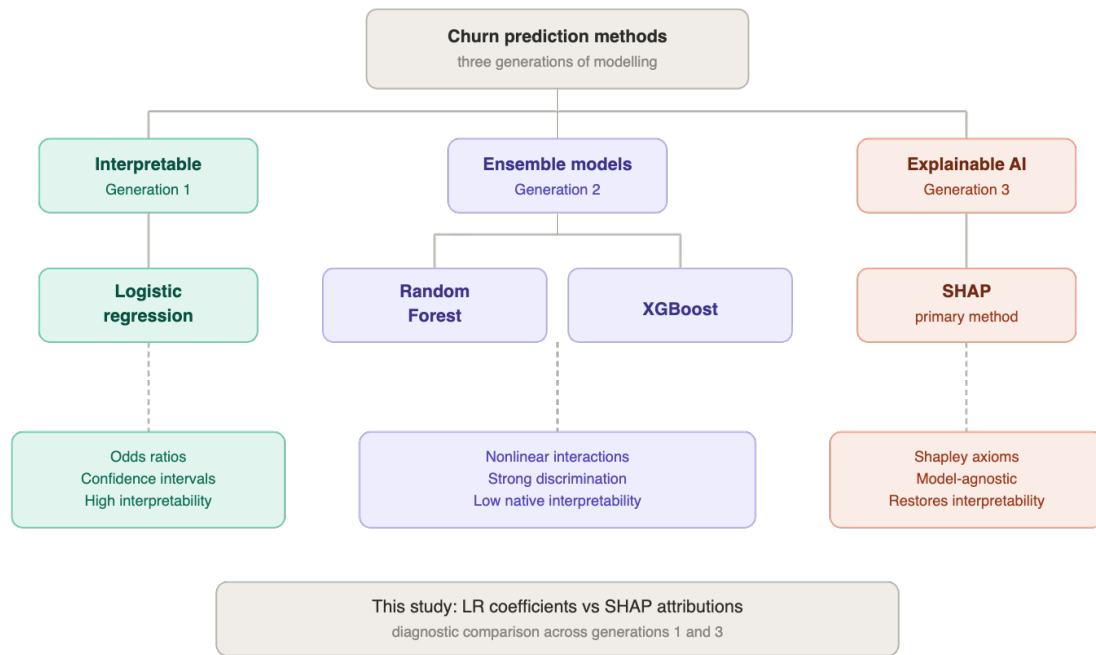


Figure 2.2: Taxonomy of churn prediction modelling approaches and the interpretability–performance trade-off.

Logistic regression remains the standard interpretable baseline. Its assumptions (linearity in the log-odds, observation independence, absence of multicollinearity) are rarely fully satisfied in telecom data, yet it remains the baseline because those assumptions are explicit, testable, and correctable. Lemmens and Croux (2006) found that while ensembles generally outperformed logistic regression on discrimination, the magnitude varied with feature engineering quality, suggesting the model’s primary limitation is functional form rather than statistical framework. Coussement and Van den Poel (2008) reached similar conclusions, noting that coefficient-based models remained preferred in operational contexts requiring individual prediction explanation.

Random Forest (Breiman, 2001) addresses the linearity constraint through bootstrap-aggregated decision trees, producing a low-variance ensemble suited to tabular churn data exhibiting skew and threshold effects. Its interpretability limitation is significant: native mean-decrease-in-impurity importance is biased toward high-cardinality and continuous

features and provides no information about effect direction (Strobl et al., 2007), motivating SHAP application.

XGBoost (Chen and Guestrin, 2016) extends gradient boosting with regularisation to reduce overfitting. Gradient boosting has achieved strong discrimination on telecom churn datasets (Ahmad et al., 2019; Ullah et al., 2019), though cross-study comparison is complicated by preprocessing and evaluation differences. Verbeke et al. (2012) provided a corrective to the performance-first paradigm: maximising AUC without accounting for cost structure and interpretability can lead to suboptimal model selection. Recent applications confirm Random Forest exceeding 91% accuracy with XAI techniques (Chang et al., 2024); XGBoost has been shown to outperform Random Forest across most metrics under upsampled imbalance (Imani et al., 2024). A notable limitation is that XGBoost's native importance scores lack the consistency property: a feature can receive lower importance even when its true contribution increases (Lundberg and Lee, 2017), motivating SHAP.

2.4 Model Evaluation in Churn Prediction

Evaluation metric selection determines which aspects of model behaviour are rewarded. The churn prediction literature has historically favoured accuracy and AUC-ROC, but neither is fully adequate for class-imbalanced, cost-asymmetric contexts. In the Iranian Churn dataset, a null classifier predicting universal retention achieves 84.35% accuracy without identifying a single churner. Hand (2009) argued accuracy is poorly suited to imbalanced classification, a view since widely accepted.

AUC-ROC addresses threshold-dependence but suffers a known optimism problem under class imbalance: because the false positive rate uses total negatives as denominator, a model that misclassifies many retained customers can still achieve high AUC-ROC when negatives are abundant. This is problematic because minority precision, which determines retention

campaign efficiency, becomes strategically more relevant than overall ranking discrimination under severe class imbalance. Davis and Goadrich (2006) demonstrated formally that PR curves provide more informative pictures of classifier performance than ROC curves when the positive class is rare, with Saito and Rehmsmeier (2015) providing supporting analysis. This study designates PR-AUC as the primary discrimination metric, retaining AUC-ROC for comparability with prior work.

Probability calibration has received limited attention in the churn literature but is essential for operational deployment. A well-calibrated model produces probability estimates that match empirical frequencies: when it assigns 30% churn probability, approximately 30% of those customers should actually churn. Niculescu-Mizil and Caruana (2005) showed that several ensemble methods produce miscalibrated outputs (Naeini et al., 2015). The Brier score (Brier, 1950; Murphy, 1973) quantifies probabilistic accuracy as mean squared error between predicted probabilities and actual outcomes. For systems deployed as probability scorers, poorly calibrated probabilities systematically distort intervention prioritisation and budget allocation, misallocating retention resources toward customers whose true risk deviates from model-reported probabilities. This study reports Brier scores and calibration curves for all three models, a combination underreported relative to its deployment importance.

2.5 Model Interpretability and Explainable AI

Retention decisions in telecoms increasingly rely on automated churn scoring, creating accountability requirements discriminative models alone cannot meet. Doshi-Velez and Kim (2017) argue that interpretability is a scientific requirement: a model that cannot explain its predictions provides no dependable means to detect when it is failing consistently.

Lipton (2018) identified three facets of interpretability: simulatability, decomposability, and algorithmic transparency. Logistic regression satisfies all three; ensemble methods satisfy

none natively. Rudin (2019) challenged the framing of interpretability and performance as inherently opposing, arguing that for well-structured tabular data interpretable models can achieve competitive performance with black-box alternatives when features are well-engineered. The argument is directly relevant: logistic regression achieves strong AUC-ROC on the Iranian dataset, raising the question of whether the interpretability cost of ensembles is justified by the performance gain, or whether both properties can be partially recovered through post-hoc explanation.

Post-hoc methods explain a trained model without modifying it. Barredo Arrieta et al. (2020) categorised these along dimensions of local versus global scope and model-agnostic versus model-specific implementation. Three approaches dominate practical application. LIME (Ribeiro et al., 2016) approximates predictions with local linear surrogates but is sensitive to the choice of perturbation sampling and inconsistent for similar observations (Molnar, 2022). Counterfactual explanations (Wachter et al., 2017) identify the smallest changes to features that would change a prediction, providing recourse at the instance level, but limited ability to interpret global behaviour. SHAP (Lundberg and Lee, 2017) grounds feature attribution in cooperative game theory's Shapley values, guaranteeing consistent, locally accurate attributions and a unified structure for local and global explanations. Sundararajan and Najmi (2020) analysed the axiomatic basis; Lundberg et al. (2020) extended the framework with TreeSHAP, enabling exact Shapley computation for tree-based models in polynomial time.

SHAP is especially appropriate to this dissertation's analytical objectives. Its consistency property enables principled comparison with logistic regression coefficients, the core diagnostic exercise in Chapter 4; its dual local-global structure supports both individual-customer explanation and the aggregate feature-importance analysis required for CLV segmentation; TreeSHAP delivers exact attribution for XGBoost in polynomial time,

avoiding the perturbation-sampling instability that affects LIME; and its theoretical guarantees make it defensible when divergences from a linear baseline are themselves the analytical focus. SHAP's principal limitations, independent feature marginalisation under correlation (Hooker et al., 2021) and the absence of causal interpretation, are addressed through the structural diagnostic framework introduced in Chapter 4. Interpretability is treated not as a descriptive output but as a diagnostic instrument for nonlinear behavioural dynamics in customer churn.

Telecom churn feature importance analyses have tended to identify usage intensity and service failure indicators among the most influential predictors, consistent with EDT (Ullah et al., 2019; Umayaparvathi and Iyakutti, 2016). A recurring limitation is that SHAP is applied to a single model in isolation, without structured comparison to an interpretable baseline. Recent applications continue this descriptive pattern: El Attar and El-Hajj (2026) achieved AUC-ROC of 0.932 on a contemporary telecom dataset identifying contract type and tenure as dominant drivers, and Bolla and Narsepalle (2026) showed SHAP provides actionable retention insights, but neither cross-references SHAP attributions against a linear baseline. This dissertation treats SHAP not as a standalone explanatory output but as a comparative diagnostic framework for identifying where nonlinear behavioural relationships diverge from linear assumptions.

2.6 Synthesis and Identified Research Gaps

The literature reveals three unresolved methodological tensions relevant to this research.

The discrimination–calibration gap. Churn studies have prioritised discrimination, particularly AUC-ROC, while neglecting probability calibration. For systems deployed as probability scorers, miscalibration distorts intervention prioritisation and budget allocation,

yet Brier scores remain underreported. This study addresses the gap by reporting calibration metrics for all three models and evaluating deployment implications.

The interpretability–performance gap. On discrimination framed as an interpretability trade-off, ensemble methods generally outperform logistic regression. This framing is incomplete; SHAP extracts explanations from black-box models but published studies use it descriptively on a single model, rather than diagnostically against an interpretable baseline, leaving the linear-assumption violations unidentified. This research addresses this gap by providing a structured comparison of SHAP versus LR, which identifies features that violate the linear log-odds assumption, explains divergences with EDT and Switching Cost Theory, and reconceptualises interpretability as a diagnostic tool rather than a descriptive output.

The methodological rigour gap. Published churn studies frequently use single train-test splits with SMOTE applied before splitting rather than within training folds, introducing leakage and performance inflation (Blagus and Lusa, 2013); feature importance is commonly reported without stability assessment. This dissertation addresses the gap through repeated stratified cross-validation, a leakage-free pipeline, and bootstrap SHAP stability analysis with 95% confidence intervals.

These gaps map directly to the research questions: the discrimination–calibration gap motivates RQ1, the interpretability–performance gap motivates RQ2, and the methodological rigour gap underlies all three. Addressing these questions requires a rigorously controlled modelling pipeline, the design of which is described in Chapter 3.

CHAPTER 3: METHODOLOGY

This chapter describes the data, preprocessing, modelling, evaluation, and interpretability procedures employed to address the four objectives and three research questions set out in Chapter 1. The methodological design prioritises three objectives simultaneously: predictive discrimination, probability reliability, and interpretability robustness. Figure 3.1 summarises the analytical pipeline.

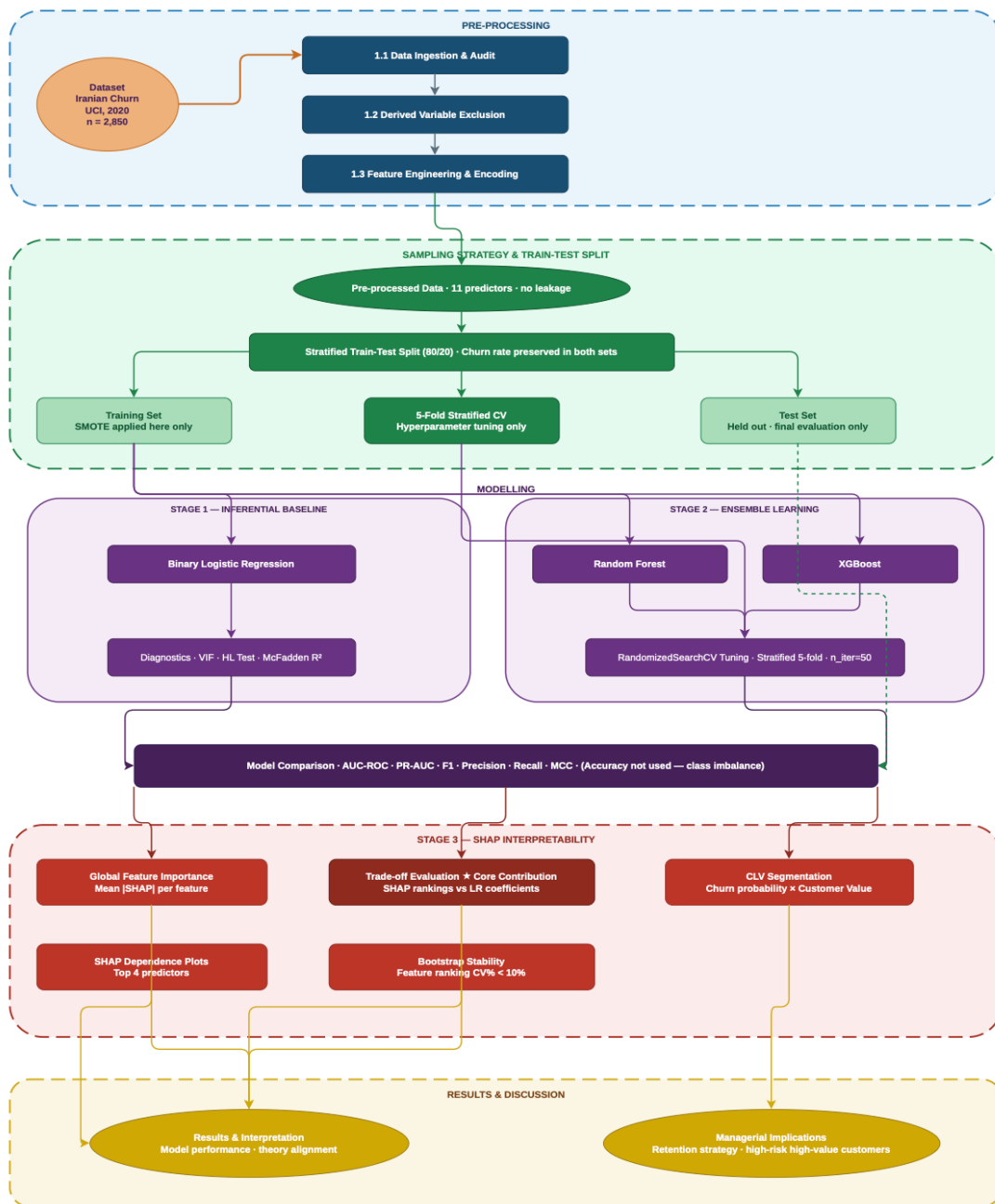


Figure 3.1: Overview of the analytical pipeline from data ingestion to business decision framework.

3.1 Research Philosophy and Design

The study adopts a positivist research philosophy, treating churn prediction as an empirical problem in which observable behavioural and contractual variables are used to estimate the probability of customer defection. The objective is comparative evaluation of predictive relationships under controlled experimental conditions (Bryman, 2016), not causal inference. Three models are evaluated against a common metric set on a shared dataset, with all analytical decisions made on the training partition and reported outcomes derived from an untouched held-out test set. Stochastic operations use a fixed random seed (`random_state = 42`) for reproducibility. Analyses were performed in Python 3.10 using scikit-learn, XGBoost, imbalanced-learn, SHAP, and statsmodels.

3.2 Dataset Description

The Iranian Telecom Churn Dataset (Jafari-Marandi et al., 2020; UCI Repository ID 563) contains twelve-month behavioural and contractual records for 3,150 telecom subscribers from a single Iranian provider. Following removal of 300 exact duplicate rows, the analytical sample comprises 2,850 observations. The target variable is a binary indicator that the customer cancelled service during the observation window. There are no post-churn fields (termination dates, cancellation flags) so all predictors are based on information prior to the churn event, so there is no target leakage from post event factors. The 15.65 percent churn rate constitutes a moderately imbalanced classification problem (He and Garcia, 2009) with 446 churners, and 2,404 retained, a 5.4:1 class imbalance. A null classifier predicting universal retention achieves 84.35% accuracy without identifying any churner, rendering raw accuracy uninformative.

The dataset includes eleven predictors retained after preprocessing, spanning four construct domains: service engagement (Frequency of Use, Seconds of Use, Frequency of SMS,

Distinct Called Numbers), service failure (Call Failures, Complains), contractual status (Tariff Plan, Subscription Length, Status), and charge behaviour (Charge Amount, Age Group). The single-provider, single-country, decade-old origin is acknowledged as a generalisability constraint, partially offset by the theoretical (non-market-specific) grounding of the behavioural mechanisms and tested directly through cross-study comparison in Section 5.5.

Attribute	Construct Domain	Definition
Call Failures	Service failure	Total number of failed call attempts experienced by the customer (count, range 0–36).
Complains	Service failure	Whether the customer filed a formal complaint during the observation window (binary: 0 = no complaint, 1 = complaint).
Subscription Length	Contractual	Total months the customer has been subscribed to the service (count, range 3–47).
Charge Amount	Charge behaviour	Customer's billing tier (ordinal: 0 = lowest tier, 9 = highest tier).
Seconds of Use	Service engagement	Total seconds of voice calls aggregated over the first nine months (count, range 0–17,090).
Frequency of Use	Service engagement	Total number of voice calls placed (count, range 0–255).
Frequency of SMS	Service engagement	Total number of text messages sent (count, range 0–522).
Distinct Called Numbers	Service engagement	Total number of unique phone numbers contacted by the customer (count, range 0–97).
Tariff Plan	Contractual	Customer's billing arrangement (binary: 1 = Pay-as-you-go, 2 = Contractual).
Status	Contractual	Account activity status at the close of the observation window (binary: 1 = Active, 2 = Non-active).
Age Group	Customer attribute	Banded customer age (ordinal: 1 = youngest age band, 5 = oldest age band).

Table 3.1: *Description of the eleven predictor variables in the analytical sample.*

3.3 Data Preprocessing and Feature Selection

Preprocessing followed a differential strategy: tree-based models, invariant to monotonic transformations, received no scaling; logistic regression required transformation to satisfy

distributional assumptions. For LR, right-skewed counts (Call Failures, Frequency of SMS) were log1p-transformed; remaining continuous variables (Subscription Length, Seconds of Use, Frequency of Use, Distinct Called Numbers) were transformed using Yeo-Johnson (Yeo and Johnson, 2000) and standardised. Binary and ordinal variables required no transformation. All transformers were fitted on training data only, preventing leakage.

Multicollinearity was assessed via Variance Inflation Factors. Frequency of Use (VIF = 17.64) and Seconds of Use (VIF = 13.52) exceed the conventional threshold of 10 and are highly correlated ($r = 0.946$). Multicollinearity primarily affects coefficient stability in logistic regression but does not affect predictive performance in any of the three models; retention of both variables therefore preserves behavioural signal without compromising validity. The regularised L1 and L2 versions gave similar signs and directions of coefficients, suggesting that multicollinearity does not greatly influence substantive conclusions.

Feature selection: The raw UCI dataset contains sixteen columns: eleven behavioural and contractual predictors, the binary target, and four excluded variables. Age (continuous) was excluded in favour of Age Group (ordinal). Customer Value was retained for segmentation analysis (Section 4.5) but excluded from prediction as it represents an outcome measure. Two derived financial metrics (FN, FP) were excluded as pre-computed cost estimates. No feature engineering was performed: all three models were trained on the same eleven original features to isolate model architecture effects from feature engineering quality.

3.4 Pre-Modelling Diagnostics

Three diagnostic checks were performed on training data prior to logistic regression estimation; tree-based models impose no distributional assumptions. At first, optimal class separation was evaluated via cross-tabulation: no perfect separation was identified; however, Complains exhibited a significant conditional churn rate (81.32% for Complains = 1

compared to 9.96% for Complains = 0). Secondly, the Box-Tidwell test (Box and Tidwell, 1962) evaluated log-odds linearity by using an $X \cdot \ln(X)$ interaction term; two predictors yielded significant results: Subscription Length ($p = 0.017$) and Frequency of SMS ($p = 0.001$), signifying nonlinearity in the log-odds. Their coefficients are interpreted as linear approximations; SHAP dependence plots in Section 4.3.2 examine the true nonlinear shapes. Third, Cook's Distance identified 143 observations (6.27%) exceeding the $4/n$ threshold with maximum value 0.022, well below 0.5, confirming no identifiable subgroup disproportionately drives the model.

3.5 Class Imbalance Handling

The 5.4:1 class imbalance was addressed using SMOTE (Chawla et al., 2002), a well-established intervention for imbalanced binary classification (He and Garcia, 2009; Krawczyk, 2016), which generates synthetic minority observations by linear interpolation between existing minority instances, expanding the training set from 2,280 to 3,846 observations at a 1:1 ratio. Applying SMOTE before splitting would allow synthetic observations to appear in both training and validation folds, introducing leakage (Blagus and Lusa, 2013); SMOTE was therefore applied inside each cross-validation fold via an imbalanced-learn Pipeline. An empirical comparison of SMOTE and SMOTE+Tomek Links (Batista et al., 2004) confirmed Tomek removal eliminated informative boundary observations rather than noise on this structurally separable dataset; SMOTE was therefore retained. Figure 3.2 illustrates the effect of SMOTE on the class distribution across the training and test partitions.

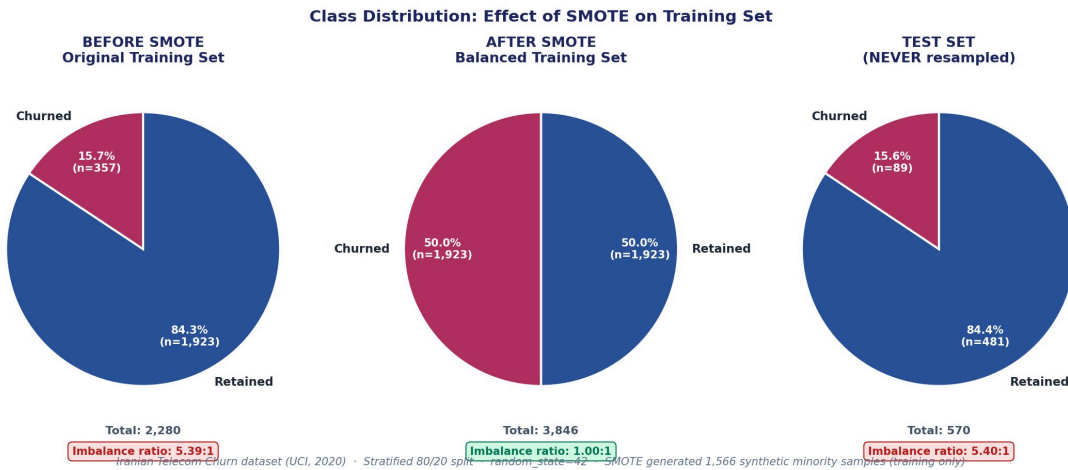


Figure 3.2: *Class distribution before and after SMOTE on the training set, and the preserved natural imbalance on the held-out test set.*

3.6 Model Development

The three selected models represent complementary modelling paradigms along the interpretability–performance spectrum. Logistic regression was retained not merely as a baseline but as an interpretable parametric reference against which nonlinear behavioural deviations identified through SHAP could be systematically evaluated; its explicit coefficient structure makes violations of linearity detectable and theoretically interpretable. Random Forest is a variance-reducing ensemble of decision trees; XGBoost is a regularised gradient boosting model capable of capturing complex nonlinear interactions.

Logistic regression estimates the probability of churn as a logistic function of a linear predictor (Equation 3.1), with the equivalent log-odds representation in Equation 3.2. The model was fitted using statsmodels for full inferential output, with a parallel scikit-learn implementation for downstream pipeline compatibility.

$$P(Y=1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_k X_k)}} \quad (3.1)$$

$P(Y=1|X)$ = predicted probability of churn; $X_1 \dots X_k$ = predictor variables; $\beta_0 \dots \beta_k$ = coefficients estimated from training data.

$$\log \left(\frac{P(Y=1)}{1-P(Y=1)} \right) = \beta_0 + \sum_{i=1}^k \beta_i X_i \quad (3.2)$$

The left-hand side is the log-odds of churn; linearity in this form is the assumption evaluated by the Box–Tidwell test.

Random Forest (Breiman, 2001) is an ensemble of decision trees, each tree is trained on a random subsample of observations and a random subset of features at each split node and aggregated by bootstrapping. Equation (3.3) formalises this, where the ensemble prediction is the simple average of all individual tree predictions. Variance is greatly reduced compared to that of individual trees by aggregating many trees, while maintaining low bias. Random Forest is well suited for datasets with nonlinear relationships and feature interactions.

$$\hat{f}(x) = \frac{1}{B} \sum_{b=1}^B T_b(x) \quad (3.3)$$

where $T_b(x)$ is the prediction of the b -th decision tree and B is the total number of trees. The averaging process reduces variance without increasing bias (Breiman, 2001).

XGBoost extends gradient boosting (Friedman, 2001) by summing contributions from K sequential trees that correct residuals of the preceding ensemble (Equation 3.4), optimised by minimising a regularised objective with L1 and L2 penalties.

$$\hat{y}_l = \sum_{k=1}^K f_k(x_i) \quad (3.4)$$

where $f_k(x_i)$ is the prediction from the k -th tree and K is the total number of boosting rounds; each tree corrects the residuals of the previous ensemble.

$$L(\phi) = \sum_i l(y_i, \hat{y}_i) \sum_k \Omega(f_k) \quad (3.5)$$

where l is the binary cross-entropy loss, $\Omega(f_k) = \gamma T + \frac{1}{2} \lambda w^2$ is the regularisation term penalising tree complexity T and leaf weights w , and ϕ denotes the full set of model parameters (Chen and Guestrin, 2016).

Both ensembles were tuned with RandomizedSearchCV (50 sampled configurations, 5-fold stratified cross validation, SMOTE in-fold, AUC-ROC as the optimisation objective). Randomised search was preferred over exhaustive grid search to explore high-dimensional spaces in a tractable way (Bergstra and Bengio, 2012). Tuning targeted tree-depth (controlling over-fitting), learning-rate (controlling generalisation stability) and L1/L2 regularisation (reducing variance). XGBoost's `scale_pos_weight` was set to 1 as the training data was already SMOTE-balanced. The held-out test set ($n=570$) was strictly separated from pre-processing, tuning, and model selection. As Cawley and Talbot (2010) note, non-nested cross-validation may have a slight optimistic bias in the CV scores, but the final reported figures are unaffected by the held-out test set. Given the study's comparative rather than benchmark-maximisation objective, repeated external test-set evaluation was prioritised over computationally intensive nested optimisation.

The three models cover the spectrum of bias-variance (Hastie et al., 2009): logistic regression (high bias, low variance), Random Forest (reduced variance by aggregation) and XGBoost

(reduced bias by sequential residual fitting with regularisation). Their comparison is a formal test of the effect of model complexity on predictive accuracy and interpretability.

3.7 Evaluation Framework

Evaluation metrics were chosen to present informative assessment of minority-class performance, rather than overall accuracy. The primary discrimination metric is the Precision-Recall AUC (PR-AUC), which is preferred to ROC-AUC for imbalanced data because it is less inflated by the large number of true negatives (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). The research also reports ROC-AUC for comparison with other works. F1-score, Matthews Correlation Coefficient (MCC), precision and recall at the default threshold are also presented in this study. MCC takes into account all four cells of the confusion matrix and is a balanced metric in the case of imbalance (Chicco and Jurman, 2020).

The Brier score (Brier, 1950) and the Hosmer-Lemeshow test (Hosmer et al., 2013) were used to evaluate calibration. Poorly calibrated models misallocate resources and skew customer priority due to threshold dependence and budget constraints, making probability calibration an operationally crucial component of retention interventions. The Hosmer-Lemeshow test is sensitive to sample size; because SMOTE inflates the training set, it tends to reject calibration even when miscalibration is minor. Calibration plots are therefore prioritised over significance tests.

$$\text{Brier Score} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2 \quad (3.6)$$

p_i = predicted churn probability for observation i ; y_i = observed binary outcome (0 = retained, 1 = churned);
 N = number of observations. A score of 0 indicates perfect calibration.

3.8 SHAP Explainability Analysis

SHAP (Lundberg and Lee, 2017) was applied to the best-performing ensemble model on its raw log-odds (margin) output, which TreeSHAP explains exactly; probability calibration is assessed separately in Section 4.2.4 and recommended for deployment. Feature attributions are generated grounded in cooperative game theory. The Shapley value ϕ_i of feature i is its average marginal contribution to the prediction across all possible feature orderings (Equation 3.7). This formulation satisfies four axioms (efficiency, symmetry, dummy, additivity) that arbitrary importance measures do not (Lundberg et al., 2020). SHAP values are on the log-odds scale, making them directly comparable to logistic regression coefficients and enabling the structured diagnostic comparison central to this study.

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (3.7)$$

where S is a subset of all features F excluding feature i , $f(S)$ is the model prediction using only features in S , and the weighting term ensures all orderings are weighted equally; ϕ_i satisfies efficiency, symmetry, dummy, and additivity (Lundberg and Lee, 2017; Shapley, 1953).

SHAP TreeExplainer was used for efficiency with tree-based models, generating global importance (mean absolute SHAP across test observations) and local per-customer attributions. Dependence plots were examined for nonlinear relationships, particularly for Subscription Length and Frequency of SMS where Box-Tidwell confirmed nonlinearity. Bootstrap stability analysis (100 resamples with replacement) tested whether SHAP rankings are population-level estimates rather than test-set artefacts; features with CV% below 10% were classified as stable. A methodological caveat applies to correlated features: Shapley values assume features can be independently marginalised, but when predictors are highly correlated (Frequency of Use and Seconds of Use, $r = 0.946$), attributions may distribute

importance in ways that do not reflect independent contributions (Hooker et al., 2021). Bootstrap stability addresses sampling variance but not this structural concern; attributions for highly correlated features should be interpreted as reflecting their joint signal.

3.9 Business Decision Framework

Translating churn probability estimates into operational retention decisions requires two components: cost-sensitive threshold optimisation and customer lifetime value segmentation. Standard threshold optimisation maximises F1, implicitly treating false positives and false negatives as equally costly. In practice, a missed churning represents the loss of their full revenue stream while an unnecessary retention offer represents only the intervention cost. A 5:1 cost ratio was adopted as the baseline, grounded in the dataset's Customer Value distribution (mean ≈ 530 units; typical intervention ≈ 100 units). The theoretical cost-optimal threshold, formalised in Equation (3.8), yields $t^* \approx 0.17$ under this ratio. Sensitivity was tested across cost ratios of 2:1, 5:1, and 10:1 (Elkan, 2001).

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{1}{1+5} \approx 0.17 \quad (3.8)$$

C_{fp} = cost of false positive (unnecessary retention offer); C_{fn} = cost of false negative (missed churning).
Substituting the 5:1 baseline cost ratio yields $t^* \approx 0.17$.

Customer Lifetime Value (CLV) segmentation combines predicted churn probability with Customer Value to build a retention priority matrix, addressing the observation that the highest risk customers are not necessarily the highest value customers and therefore not the highest return retention targets. The matrix identifies mid-to-high value customers at early warning stage (first complaint or declining usage) as the optimal target segment.

3.10 Robustness Validation

This study carried out six robustness checks to validate single-split results over data partitions: (i) 5-fold stratified cross-validation with in-fold SMOTE to ensure split-independence; (ii) learning curves to assess bias-variance and data sufficiency; (iii) threshold sensitivity to quantify F1, precision and recall trade-offs across thresholds 0.05–0.95; (iv) SHAP bootstrap stability (100 resamples and 10% CV% threshold) to validate importance rankings; (v) cost-sensitive analysis to test sensitivity to FN/FP cost ratios across 2:1, 5:1 and 10:1 scenarios; (vi) feature selection robustness by re-training all models under three regimes (full eleven, VIF-reduced ten and SHAP top-six) to confirm comparative conclusions.

3.11 Ethical, Legal and Social Considerations

The study uses a publicly available secondary dataset from the UCI Machine Learning Repository containing no personal identifiers. No primary data collection was conducted and no human participants were involved; formal ethical approval was therefore not required under De Montfort University guidelines for secondary data analysis.

Broader ethical, legal, and social considerations nonetheless apply to the deployment context this dissertation addresses. Ethically, automated churn-based targeting may disadvantage vulnerable customers whose low usage reflects accessibility or socioeconomic constraints rather than disengagement, and opaque predictive systems may reduce transparency in commercial interactions. Legally, any production deployment would require compliance with the UK General Data Protection Regulation, particularly Article 22 provisions governing automated profiling and customer data processing. Socially, retention systems that prioritise high-value customers may reinforce uneven treatment across the customer base.

Several of these concerns are partially addressed by the analytical choices made in this study: the SHAP explainability layer supports the transparency requirements that data-protection

regulators cite as essential for automated decision systems, and the explicit documentation of model assumptions, decision thresholds, and targeting rules provides a traceable audit record. Full production deployment would additionally require human oversight of individual targeting decisions, periodic fairness auditing by customer subgroup, and transparent data governance policies.

CHAPTER 4: RESULTS

4.1 Dataset Overview

The analytical sample includes 2,850 observations from the Iranian Telecom Churn Dataset after the removal of 300 exact duplicate rows. The test set consists of 570 observations (20%) with 89 churners and 481 retained. The natural class distribution (15.65% overall; 15.61% in the test partition) is preserved. The classes are imbalanced (5.4:1), so raw accuracy is not informative. PR-AUC is subsequently employed as the main evaluation metric.

4.2 Model Performance and Behaviour

This section presents a structured evaluation of model performance across discrimination, calibration, and classification behaviour. Table 4.1 presents the comparative performance on the held-out test set. Bold values indicate the best result per column.

Model	PR-AUC	AUC-ROC	F1	Precision	Recall	MCC	Brier
Logistic Regression	0.752	0.924	0.633	0.503	0.854	0.574	0.112
Random Forest	0.874	0.978	0.800	0.694	0.944	0.770	0.054
XGBoost	0.922	0.983	0.812	0.726	0.921	0.780	0.050

Table 4.1: Comparative performance of logistic regression, Random Forest, and XGBoost on the held-out test set.

4.2.1 Overall Performance Comparison

XGBoost yields the best results for most metrics (PR-AUC = 0.922, AUC-ROC = 0.983, F1 = 0.812, MCC = 0.780, Brier = 0.050). Random Forest achieves the highest recall (0.955 compared to 0.921 for XGBoost), indicating that its variance reduction mechanism favours sensitivity over precision. Logistic regression performs well on AUC-ROC (0.924, strong rank-ordering), but poorly on precision (0.503), meaning that half of all flagged customers

would not actually churn. The F1 gaps between logistic regression (0.633) and XGBoost (0.812) are 17.9 percentage points.

4.2.2 Sensitivity Analysis: Ridge LR and Influential Observations

Ridge logistic regression ($C = 0.5$, selected via 5-fold cross-validation) produced coefficient estimates that were stable relative to the standard model across all 11 predictors, with nine of eleven features rated Stable (OR change $< 10\%$). Seconds of Use exhibited a moderate reduction in odds ratio under regularisation (OR change -18.1%), consistent with its high VIF of 13.52 and collinearity with Frequency of Use. This confirms that the standard LR coefficients are not materially distorted by multicollinearity. Cook's Distance analysis identified 143 influential observations (6.3%), with a maximum of 0.022, well below the 0.5 threshold, confirming LR coefficients reflect population-level patterns rather than individual outliers.

4.2.3 Discrimination Analysis

While AUC-ROC is reported for comparability with earlier work, PR-AUC is prioritised since it directly measures minority-class discrimination without being inflated by the majority class. The PR curves show more distinct separation than the ROC curves, which cluster closely together (0.924-0.983). LR achieves a PR-AUC of 0.752, Random Forest 0.855, and XGBoost 0.922. XGBoost maintains substantially higher precision at recall levels above 0.70, reflecting better resolution of the churn probability distribution near the decision boundary.

Model Comparison — ROC & Precision-Recall Curves

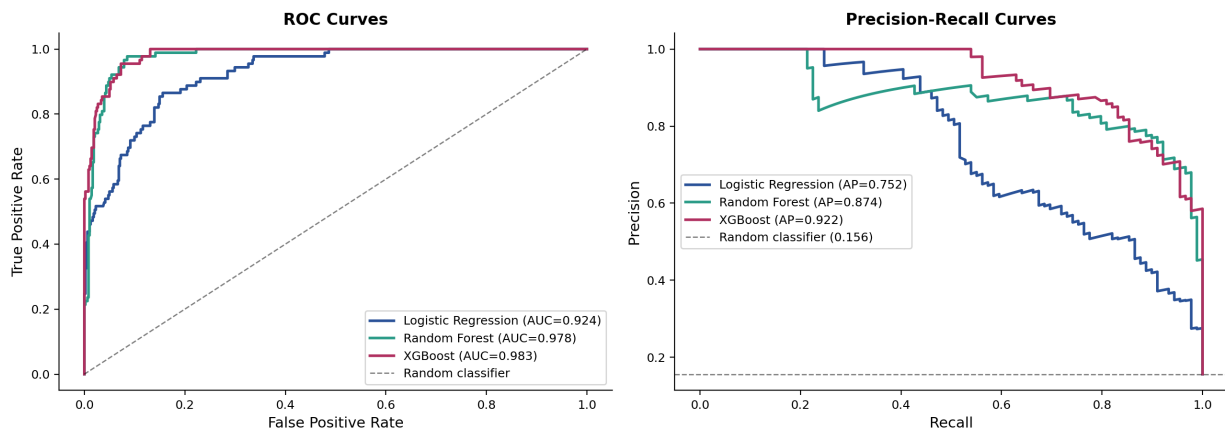


Figure 4.1: Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves for logistic regression, Random Forest, and XGBoost.

The PR-AUC gap is consistent with XGBoost’s capacity to capture nonlinear interactions not representable in logistic regression. Box-Tidwell diagnostics showed that the features Subscription Length ($p = 0.017$) and Frequency of SMS ($p = 0.001$) were nonlinear, meaning that the linear boundary systematically misclassifies customers where the probability of churn depends on nonlinear combinations of features.

4.2.4 Calibration Analysis

Brier scores indicate XGBoost (0.050) and Random Forest (0.054) produce more reliable probability estimates than logistic regression (0.112). The Hosmer-Lemeshow test rejects perfect calibration under the SMOTE training set (HL = 85.38, $p < 0.001$) but falls to marginal rejection on the original data (HL = 16.99, $p = 0.030$). As noted in Chapter 3, the test is sensitive to sample size, and the SMOTE-inflated training set overstates miscalibration. The marginal rejection on original data suggests residual calibration imperfection rather than structural misspecification.

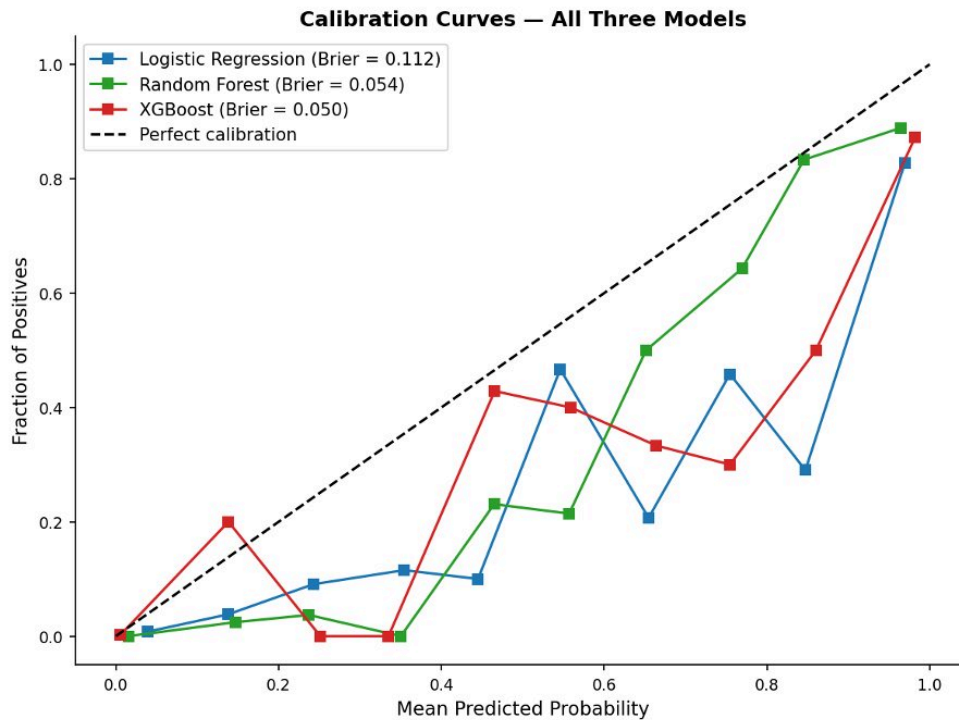


Figure 4.2: Calibration curves for logistic regression, Random Forest, and XGBoost, comparing predicted churn probabilities with observed outcomes.

As illustrated by Figure 4.2, logistic regression deviates from the diagonal at higher predicted probabilities, thus demonstrating systematic overstatement of churn certainty for the most-flagged customers. Ensemble models follow the diagonal more closely; XGBoost appears a bit overconfident at predicted probabilities > 0.7 .

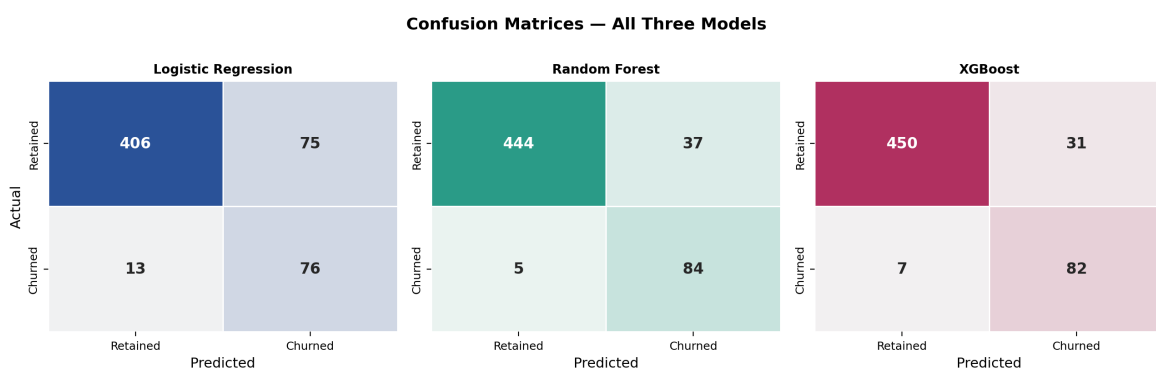


Figure 4.3: Confusion matrices for logistic regression, Random Forest, and XGBoost at the default classification threshold ($t = 0.50$).

As shown in Figure 4.3, at $t = 0.50$, logistic regression correctly identifies 76 of 89 churners, but produces 75 false positives. XGBoost identifies 82 churners with 31 false positives, a

59% reduction in wasted interventions. At the default threshold, all three models accept missed churners to reduce false positive interventions. These results should be interpreted in the context of a single-provider dataset and a non-nested validation framework, which may introduce mild optimistic bias in performance estimates (Cawley and Talbot, 2010).

4.3 SHAP Interpretability Analysis

4.3.1 Global Feature Importance

Table 4.2 shows global feature priority ranking based on mean absolute SHAP values from the XGBoost model for all 570 observations in the test set with bootstrap stability estimates from 100 resamples.

Rank	Feature	Mean SHAP	Direction	CV%	Theoretical Link
1	Frequency of Use	2.898	Negative	2.9%	Engagement decline (SCT)
2	Call Failure	1.637	Positive	2.9%	Service failure (EDT)
3	Seconds of Use	1.437	Negative	2.9%	Engagement decline (SCT)
4	Frequency of SMS	1.317	Negative	3.2%	Engagement decline
5	Charge Amount	1.107	Negative	4.7%	Pricing sensitivity (SCT)
6	Subscription Length	0.993	Nonlinear	4.3%	Tenure protection (SCT)
7	Complains	0.648	Positive	7.5%	Disconfirmation (EDT)
8	Distinct Called Numbers	0.588	Negative	3.0%	Network engagement
9	Status	0.454	Positive	1.7%	Contractual status (SCT)
10	Age Group	0.422	Negative	4.1%	Demographic
11	Tariff Plan	0.006	Negative	8.0%	Plan type (SCT)

Table 4.2: Global SHAP feature importance ranking based on mean absolute Shapley values ($n = 570$ test observations, 100 bootstrap resamples).

Frequency of Use dominates with mean |SHAP| of 2.898. Three of the top four features (Frequency of Use, Seconds of Use, Frequency of SMS) capture engagement decline. Call Failure ranks second (1.637), representing service quality failures. Subscription Length at rank 6 exhibits a nonlinear effect examined in Section 4.3.2. Tariff Plan ranks last (0.006) with negligible marginal contribution. A caveat applies for correlated features: Frequency of Use and Seconds of Use ($r = 0.946$, $VIF > 13$) may distribute importance in ways that do not reflect independent contributions (Hooker et al., 2021). Bootstrap stability confirms sampling robustness but does not address this structural concern; attributions for these features should be interpreted as their joint engagement signal.

4.3.2 SHAP Dependence Plots

The SHAP beeswarm summary plot, displaying the distribution of SHAP values for each feature across all test observations is shown next. The horizontal spread indicates the range of each feature's contribution to individual predictions, while colour encodes the raw feature value. Frequency of Use shows a left-skewed SHAP distribution: low usage customers receive large positive SHAP values (increased churn risk) while high usage customers receive strongly negative values (reduced churn risk).

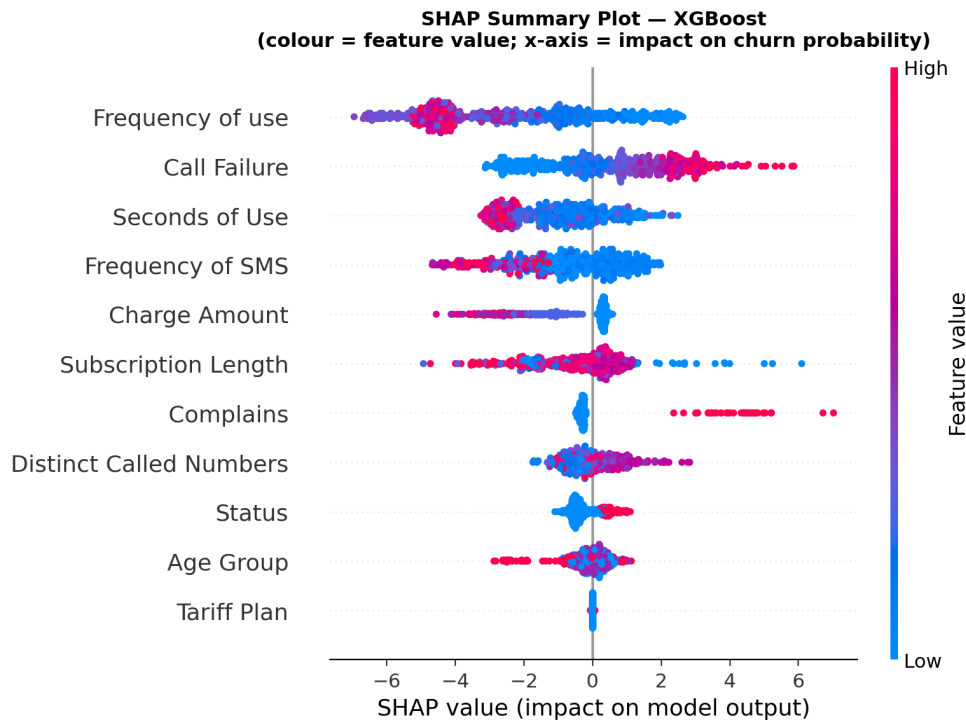


Figure 4.4: *SHAP beeswarm summary plot for XGBoost (colour = feature value; x-axis = impact on churn probability).*

Figure 4.5 presents SHAP dependence plots for the top four features by mean $|\text{SHAP}|$. Frequency of Use shows a steeply monotonic negative relationship, with the largest risk reductions concentrated among the most disengaged customers. Call Failure shows a steep positive gradient at moderate-to-high counts. Seconds of Use mirrors the usage pattern, while Frequency of SMS shows an equivalent negative relationship.

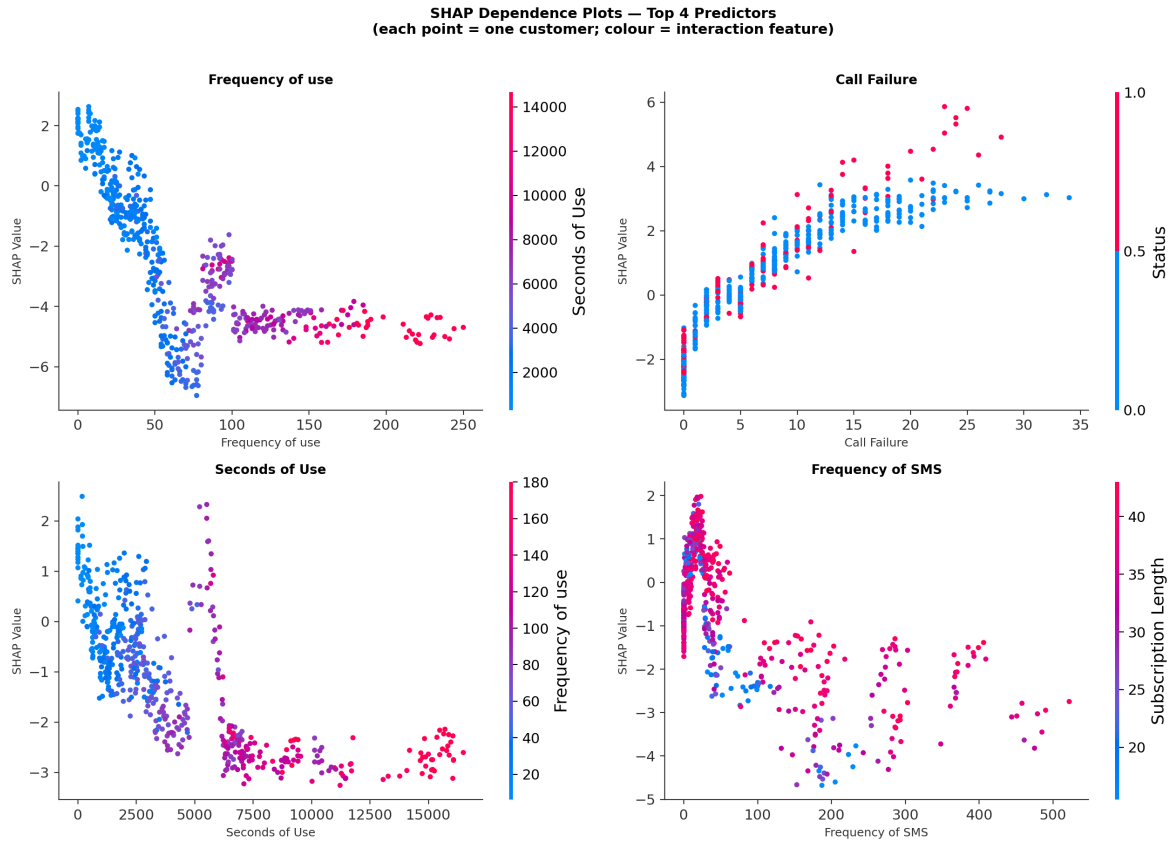


Figure 4.5: *SHAP dependence plots for the top four predictors (each point = one customer; colour = interaction feature).*

4.3.3 Bootstrap Stability

All 11 features meet the stability criterion of CV% below 10%. The top-ranked features (Frequency of Use, Call Failure, and Seconds of Use, all CV% = 2.9%) are exceptionally stable, with 95% CIs that do not overlap any lower-ranked feature. Complains, despite its lower mean |SHAP| of 0.648, remains stable at CV% = 7.5%. Tariff Plan shows the highest variability (CV% = 8.0%), consistent with its near-zero importance value where small absolute fluctuations produce relatively high coefficient of variation.

4.3.4 Local Attribution Case Studies

For Customer A ($P(\text{churn}) = 0.884$, churned), Call Failure (raw = 14, +4.13 log-odds) and Non-Active status (+0.99) drive the prediction upward, partially offset by moderate usage; net displacement +2.04 from base value. For Customer B ($P = 0.001$, retained), high

Frequency of Use (−3.21), high Charge Amount (−2.85), long Subscription Length (−1.71), and Seconds of Use (−1.55) combine to produce −11.39 log-odds against only +4.14 positive, yielding −7.26 net deflection. For borderline Customer C ($P = 0.490$, retained but misclassified at $t = 0.5$), small opposing contributions (+2.15 vs −2.18) produce a near-symmetric profile, the form of appropriately hedged prediction that a calibrated probability model produces when evidence is mixed. This per-customer decomposition is the explanation form that regulatory frameworks and retention operatives require at the individual decision level.

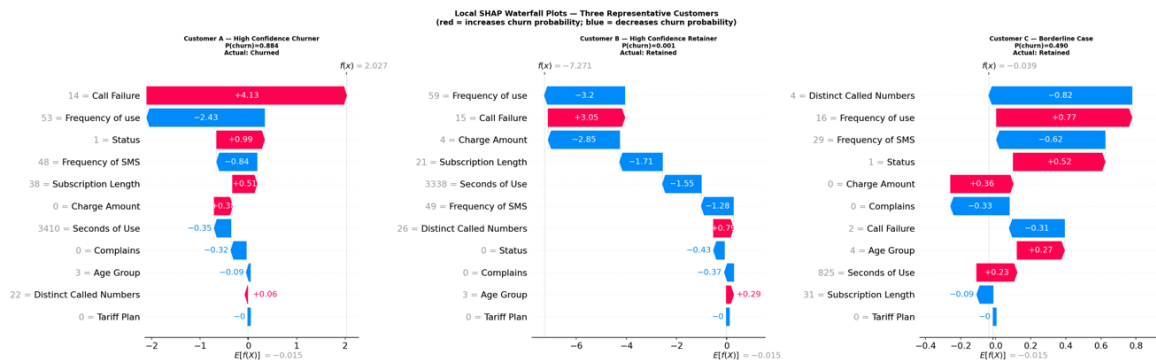


Figure 4.6: Local SHAP waterfall plots for three representative test-set customers (A: high-confidence churner; B: high-confidence retainer; C: borderline case) under XGBoost.

4.4 Logistic Regression vs SHAP Comparison

Table 4.3 compares logistic regression coefficient rankings against SHAP feature importance rankings for XGBoost across all 11 predictors; theoretical interpretation follows in Chapter 5.

Feature	LR Rank	Odds Ratio	SHAP Rank	Mean SHAP	Agreement	Interpretation
Frequency of Use	1	0.135	1	2.898	✓ Strong	Consistent: engagement decline drives churn
Call Failure	2	3.651	2	1.637	✓ Strong	Consistent: service failure operationalises EDT
Seconds of Use	5	2.574	3	1.437	~ Moderate	LR underweights; SHAP captures usage intensity
Frequency of SMS	7	0.596	4	1.317	~ Moderate	LR underweights; messaging as engagement signal
Charge Amount	4	0.382	5	1.107	✓ Strong	Consistent: lower charge signals disengagement
Subscription Length	10	0.793	6	0.993	✗ Diverges	LR misses nonlinear tenure-protection effect
Complains	3	2.989	7	0.648	✗ Diverges	LR captures local OR; SHAP diluted by non-complainers
Distinct Called Numbers	9	0.730	8	0.588	✓ Strong	Consistent: network breadth as retention signal
Status	6	1.875	9	0.454	~ Moderate	LR weights contractual status more heavily
Age Group	11	0.855	10	0.422	✓ Strong	Consistent: marginal demographic effect
Tariff Plan	8	0.685	11	0.006	~ Moderate	Near-zero SHAP; LR finds weak linear signal

Table 4.3: Comparison between logistic regression coefficient rankings and SHAP feature importance rankings for XGBoost.

The comparison yields a 5-4-2 distribution: five features have strong agreement (rank difference of ≤ 1), four moderate agreement (2–3), and two substantive divergence (≥ 4). Both linear and nonlinear models have identified substantially the same dominant behavioural drivers while differing in how they represent specific relationships. Frequency of Use ranks first under both (LR OR = 0.135; SHAP = 2.898), and Call Failure ranks second (LR OR = 3.651; SHAP = 1.637). Four features show moderate agreement: Seconds of Use

(LR 5 $\rightarrow$ SHAP 3), Frequency of SMS (7 $\rightarrow$ 4), Status, and Tariff Plan (8 $\rightarrow$ 11, near-zero SHAP indicating its apparent LR association does not survive the ensemble's more flexible representation).

Subscription Length presents the largest rank divergence. LR ranked it tenth, with an odds ratio (OR = 0.793, $p < 0.001$) close to 1, indicating a weak protective effect. With a mean $|\text{SHAP}|$ of 0.993, it ranks sixth by SHAP. The dependence plot shows a U-shaped trend: churn probability is high at very short tenures, decreases in the middle tenure ranges, and increases again at long tenures, a pattern that is theoretically consistent with a vulnerability in the renewal periods under Switching Cost Theory, even though the dataset does not directly observe contract dates. Logistic regression, constrained to fit a single linear coefficient to this U-shape, averages across the curve and recovers only the small net effect because the opposing ends partially cancel.

Complains presents a divergence of different character. Under LR it ranks third (OR = 2.989); under SHAP seventh (0.648). The divergence reflects different aggregation conventions. The LR odds ratio is a conditional estimate: the multiplicative change in odds for a customer who transitions from non-complainant to complainant, holding other variables constant. SHAP's mean $|\text{SHAP}|$ aggregates absolute contributions across all 570 observations, 82.6% of whom have no complaint on record. Among complainers, SHAP contributions are high, consistent with the LR odds ratio; the population-level average is diluted by the large non-complaining majority.

4.5 Business Decision Framework

4.5.1 Threshold Optimisation

Table 4.4 shows a comparison of XGBoost with the default threshold ($t = 0.50$) and the empirically cost-optimal threshold at a cost ratio of 5:1 ($t = 0.39$). The theoretical threshold of

Elkan (2001) is $t^* \approx 0.17$, when assuming perfect calibration. The empirical optimum is higher because XGBoost exhibits mild overconfidence above predicted probability 0.7; this is a well-documented pattern in high-capacity classifiers (Guo et al., 2017), and the empirical threshold absorbs this calibration imperfection. In practice, the empirical threshold is preferred for deployment, and periodic isotonic recalibration (Zadrozny and Elkan, 2002; Niculescu-Mizil and Caruana, 2005) is recommended, in line with calibration improvements reported by El Attar and El-Hajj (2026) for tree-based ensembles.

Threshold	Precision	Recall	F1	TP	FP
$t = 0.50$ (default)	0.726	0.921	0.812	82/89	31
$t = 0.39$ (cost-optimal, 5:1)	0.708	0.955	0.813	85/89	35

Table 4.4: XGBoost performance at the default classification threshold ($t = 0.50$) and the empirically cost-optimal threshold under a 5:1 false-negative to false-positive cost ratio ($t = 0.39$).

Reducing the threshold increases the recall from 0.921 to 0.955, recovering 3 additional churners (from 82 to 85 of 89). The precision drops slightly from 0.726 to 0.708 and the F1 remains at 0.813. The directional recommendation is robust for cost ratios of 2:1, 5:1 and 10:1.

4.5.2 CLV Segmentation Matrix

Table 4.5 presents the retention priority matrix combining predicted churn risk with Customer Value quartile, using three risk bands: Low Risk (predicted churn probability < 0.30), Medium Risk (0.30 to 0.70), and High Risk (> 0.70). The corresponding visualisation is shown after the table.

Risk Segment	Value Quartile	N	Avg Churn Prob	Avg CLV	Actual Churners
Low Risk	Q1 (Low)	74	0.047	75.14	1
Low Risk	Q2 (Mid-Low)	100	0.024	175.26	2
Low Risk	Q3 (Mid-High)	128	0.006	415.36	0
Low Risk	Q4 (High)	142	0.002	1,233.40	1
Medium Risk	Q1 (Low)	11	0.453	71.09	3
Medium Risk	Q2 (Mid-Low)	6	0.447	166.71	2
Medium Risk	Q3 (Mid-High)	1	0.389	403.38	0
High Risk	Q1 (Low)	58	0.923	53.35	41
High Risk	Q2 (Mid-Low)	36	0.920	161.98	25
High Risk	Q3 (Mid-High)	13	0.966	327.01	13
High Risk	Q4 (High)	1	0.933	978.39	1

Table 4.5: Customer segmentation matrix combining XGBoost churn risk classifications and Customer Value quartiles.

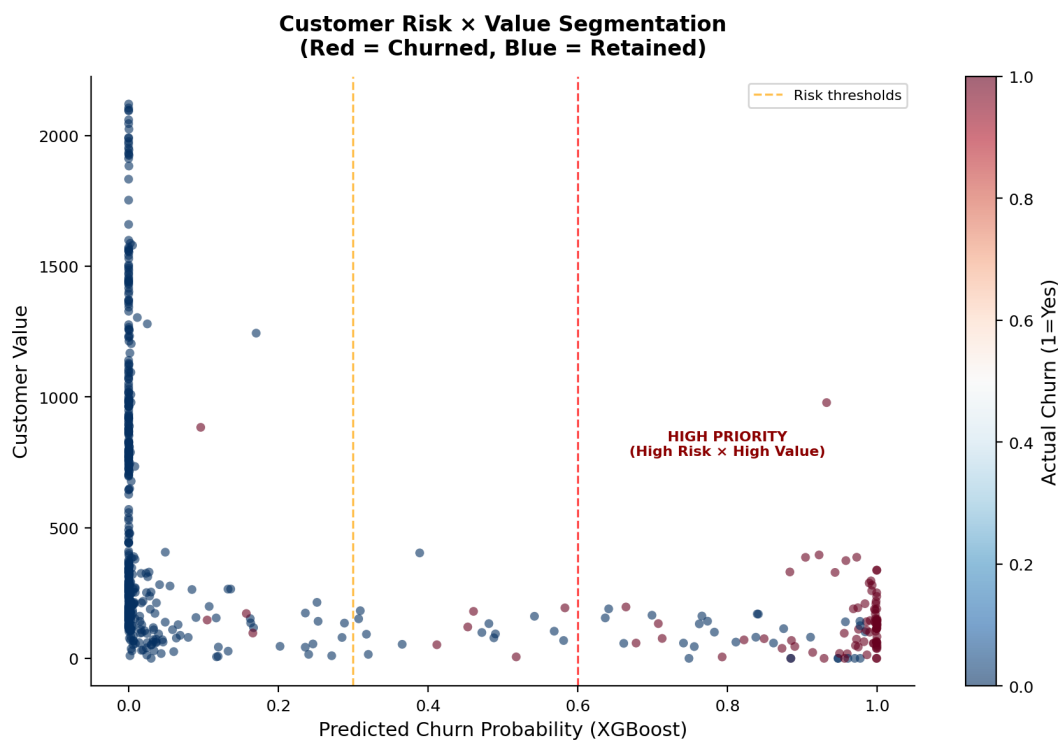


Figure 4.7: Customer segmentation scatter plot combining predicted churn probability and Customer Value.

The CLV segmentation reveals an inverse relationship between churn risk and customer value. The 58 high-risk customers in Q1 (average CLV = 53 units) account for 41 churners but minimal revenue exposure; the 49 high-risk customers in Q2 and Q3 ($n = 36$ and 13 , average CLV 162 and 327) account for 38 churners and substantially greater revenue at risk; the 142 low-risk customers in Q4 (average CLV = 1,233) have minimal churn risk (average probability = 0.002) but would generate the largest revenue loss if they churned.

4.6 Robustness Validation

Cross-validation consistency: Five-fold CV AUC-ROC estimates (LR: 0.929 ± 0.012 ; RF: 0.978 ± 0.007 ; XGB: 0.983 ± 0.006) closely match single-split results, confirming that the observed performance hierarchy is not an artefact of the specific data partition. Extended 5×5 repeated CV (XGBoost: 0.985 ± 0.004) reduces variance further.

Learning curves: Logistic regression approaches almost identical training and validation curves (gap ≈ 0.002), suggesting that the model has achieved the limit set by its linearity constraint. Ensemble models display near-perfect training AUC-ROC (≈ 1.0) but stable validation performance, indicating that regularisation effectively restricts memorisation.

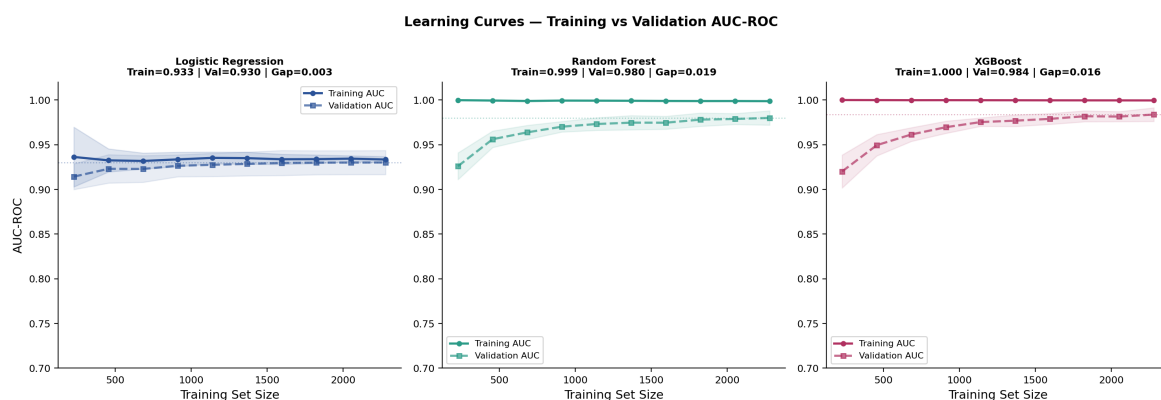


Figure 4.8: Learning curves for logistic regression, Random Forest, and XGBoost, showing training and cross-validated AUC-ROC as a function of training set size.

The AUC-ROC sums for SMOTE and SMOTE+Tomek are very similar for all three models (2.885 vs 2.888) indicating that the hierarchy of performance is not an artefact of resampling.

Feature regime comparison: Model performance under three feature regimes is presented in Table 4.6.

Feature Set	LR AUC	LR F1	RF AUC	RF F1	XGB AUC	XGB F1
Full (11 features)	0.924	0.633	0.978	0.800	0.983	0.812
VIF-reduced (10)	0.923	0.625	0.972	0.786	0.980	0.792
SHAP top-6	0.835	0.508	0.946	0.733	0.947	0.717

Table 4.6: Model performance under three feature selection regimes on the held-out test set.

The VIF-reduced and SHAP top-6 regimes provide minimal AUC decreases (LR: -0.001 to -8.9 pp; XGBoost: -0.003 to -3.6 pp), demonstrating that the ensemble advantage is a structural aspect of model flexibility and not an artefact of the particular predictors included.

Threshold sensitivity: F1-optimal thresholds (LR: 0.76, RF: 0.70, XGBoost: 0.87) are all substantially above the default of 0.50, demonstrating that the default threshold is suboptimal even by the symmetric F1 criterion. The XGboost F1 surface is flat across this region, so F1 is effectively unchanged between 0.39 and 5.0; the high F1-optimum (0.87) maximises F1 by trading recall for precision, the opposite direction to the cost objective, which is why the deployed cost-optimal threshold (0.39) is preferred.

4.7 Deployment Simulation

To translate per-customer predictive performance into operational impact estimates, a deployment simulation was conducted on the 570-observation test set comparing four retention strategies. Each active strategy targets the top 28 customers, equivalent to a 5 percent intervention budget. Intervention cost is set at 100 units per customer, approximately one-fifth of mean customer value and consistent with the 5:1 false-negative to false-positive cost ratio used in threshold optimisation, giving a total intervention budget of 2,800 units.

Intervention success probability is set at 0.30, consistent with conservative estimates in the retention literature (Verbeke et al., 2012). Expected retained value is calculated as:

$$\text{Expected Retained Value} = (\sum CV_i \text{ among correctly targeted churners}) \times p_{\text{success}}$$

The simulation is illustrative rather than predictive and compares strategic efficiency under consistent assumptions rather than estimating financial outcomes.

Strategy	Customers contacted	True churners flagged	Successful saves	Expected retained value (units)	Intervention cost (units)	Net return (units)
A: No intervention	0	0	0	0	0	0
B: Random targeting	28	≈ 4	≈ 1	≈ 254	2,800	≈ -2,546
C: Highest-risk-first	28	≈ 28	≈ 8	1,027	2,800	-1,773
D: CLV-weighted (this study)	28	≈ 22	≈ 7	2,256	2,800	-544

Table 4.7: Deployment simulation comparing retention strategies under fixed-budget intervention assumptions (n = 570; intervention budget = 2,800 units; 28 interventions at 5 percent of customer base; success probability = 0.30).

Strategy A (no intervention) loses all 89 churners and the corresponding revenue at risk (≈ 18,140 units). Strategy B (random targeting) reaches ≈ 4 churners and recovers ≈ 254 units. Strategy C (highest-risk-first) targets the top 28 customers by predicted churn probability, all of whom are actual churners, producing ≈ 8 expected saves and 1,027 units retained, but concentrates effort on the lowest-value cohort of the high-risk segment. Strategy D, the CLV-weighted framework, ranks customers by $P \times CV$; the top 28 contain ≈ 22 actual churners with substantially higher average value. Expected saves ≈ 7; expected retained value ≈ 2,256 units, a 2.20× uplift over highest-risk-first despite fewer churners saved.

The simulation shows that maximising saved churners is not the same as maximising expected retained value. Strategy C catches more churners individually. Strategy D keeps

much more value for each unit of the intervention budget, and reduces net loss by 69.4 percent: net return for Strategy D is -544 units versus -1,773 units for Strategy C. Both strategies remain net negative under the conservative 100-unit intervention cost, reflecting an intervention cost that exceeds expected per-customer revenue retained at the 30 percent success rate. The $2.20\times$ figure therefore represents a relative improvement in retained-revenue efficiency under equivalent intervention budget rather than guaranteed positive profitability. For operators with lower intervention costs typical of automated or digital retention channels (50 units or less), Strategy D converts this relative uplift into positive net retained value, while Strategy C requires intervention costs below 20 units to achieve the same. The directional recommendation is thus robust: CLV-weighted targeting outperforms highest-risk-first targeting across realistic cost structures.

4.8 Chapter Summary

The principal findings of this chapter map to the research questions as follows.

RQ1 (Performance): XGBoost achieves the strongest performance across all primary metrics (PR-AUC = 0.922, F1 = 0.812, Brier = 0.050). The performance hierarchy (LR < RF < XGBoost) is consistent across 5×5 repeated cross-validation and three alternative feature regimes.

RQ2 (Interpretability): Five of eleven features show strong agreement between LR coefficients and SHAP rankings; four show moderate agreement. Two diverge substantively: Subscription Length rises from LR rank 10 to SHAP rank 6 due to a U-shaped nonlinear effect; Complains falls from LR rank 3 to SHAP rank 7 due to differing aggregation conventions across a low base-rate feature.

RQ3 (Business decisions): The empirically cost-optimal threshold ($t = 0.39$ under the 5:1 ratio; theoretical Elkan threshold $t^* \approx 0.17$) increases recall from 0.921 to 0.955 while

preserving F1. CLV segmentation reveals that highest-risk customers are disproportionately low-value; deployment simulation confirms that CLV-weighted targeting delivers 2.20× the revenue retention of highest-risk-first targeting at equivalent intervention budget.

CHAPTER 5: DISCUSSION

5.1 Synthesis of Key Findings

The central argument of this dissertation is that the interpretability–performance trade-off in telecom churn prediction is less absolute than the framing dominant in the literature implies. Ensemble methods outperform logistic regression substantially and consistently within this dataset, but the mechanisms underlying that advantage are recoverable through structured post-hoc analysis.

SHAP analysis reveals that XGBoost and logistic regression have largely learned the same behavioural story: engagement decline and service failure are the primary churn drivers under both linear and nonlinear assumptions. Where they diverge, most clearly on Subscription Length and Complains, the divergence is theoretically interpretable, pointing to structural limitations of the linear model with direct operational consequences. Performance and interpretability are complementary analytical instruments rather than competing objectives.

5.2 Model Performance and Behaviour

The performance hierarchy ($LR < RF < XGBoost$ across all primary metrics) is interpretable through the bias–variance frame (Hastie et al., 2009). Logistic regression sits at the high-bias, low-variance end: its linearity constraint produces stable coefficients that faithfully capture population-level associations, but it underestimates churn probability whenever risk depends on feature combinations that interact non-linearly. Random Forest reduces variance via bootstrap aggregation, lifting F1 by 16.7 percentage points over LR and retaining the highest recall. That profile suits contexts where missing a churner is the dominant cost. XGBoost’s advantage of 17.9 F1 points over LR comes from its capacity to jointly model the convergent signals of usage decline, call failure, and reduced messaging: a combination the additive log-

odds form cannot represent without violating the separability assumptions that Box–Tidwell diagnostics show are not met here.

Calibration partially qualifies this picture. Ensemble Brier scores (0.050 and 0.054) are lower than LR's (0.112), but XGBoost shows slight overconfidence above predicted probability 0.7. This does not materially affect the threshold decisions in Section 4.5. It does warrant monitoring in high-stakes deployment, where probability precision at the upper tail directly affects cost–benefit calculations for individual retention offers.

In direct answer to RQ1: ensemble methods substantially outperform logistic regression on both discrimination and calibration. The advantage reflects the capacity of gradient-boosted ensembles to capture feature interactions that the LR functional form cannot represent. It holds across repeated cross-validation and three alternative feature regimes.

5.3 Interpretability and Behavioural Insights

The SHAP analysis reveals a picture of churn behaviour consistent with theoretical expectations but more subtle than the linear model captures. Frequency of Use dominates under both frameworks (mean $|\text{SHAP}| = 2.898$), consistent with Switching Cost Theory: progressive disengagement signals the erosion of integration costs that suppress switching. Three of the top four SHAP features represent usage engagement. Call Failure at rank 2 is consistent with Expectation-Disconfirmation Theory: repeated technical failures accumulate as negative disconfirmation events that independently elevate defection probability.

Subscription Length rises from LR rank 10 to SHAP rank 6 because the linear coefficient, constrained to summarise a U-shaped tenure-protection effect with a single slope, averages across a curve whose opposing ends partially cancel. The SHAP dependence plot reveals the complete non-linear pattern: risk is high at very short tenures, declines in mid-contract periods as switching costs accumulate, and then rises again as contracts near renewal, in line

with the Switching Cost Theory prediction (Jones et al., 2000). Operationally, this means that a retention system relying solely on LR coefficients would systematically under-prioritise customers close to contract renewal, the segment where proactive intervention is most effective, as the structural barrier to switching is about to diminish.

The Complains divergence is analytically distinct. The LR odds ratio captures a strong conditional effect ($OR = 2.989$) while SHAP's population-level average is diluted by the 82.6% of observations without a complaint on record. Both estimates are valid: LR captures the conditional severity of a complaint as a discrete Expectation-Disconfirmation event (Oliver, 1980); SHAP reflects its low base rate. A retention strategy relying on complaint flags alone reaches only a fraction of at-risk customers, while the broader usage-decline signal captured by SHAP provides more efficient population coverage. Among complainers, individual SHAP contributions remain high and consistent with the LR odds ratio: a first complaint should trigger immediate intervention even though it does not dominate the population-level ranking.

Together, these two divergences position SHAP as more than a description of model behaviour. It is a diagnostic that locates the points at which the linear specification distorts interpretation. Logistic regression, in turn, functions not only as an interpretable baseline but as a diagnostic foil. The 5–4–2 agreement distribution indicates the LR model has captured the dominant structure correctly; SHAP refines rather than overturns its core conclusions. The theoretical alignments are consistent with, but do not constitute proof of, the causal pathways posited by EDT and SCT.

In direct answer to RQ2: SHAP attributions broadly confirm the LR hierarchy, with five of eleven features in strong agreement. The two substantive divergences (Subscription Length, Complains) flag the precise points at which the linear baseline misrepresents the underlying behaviour, delivering interpretive value that performance metrics alone cannot.

5.4 Business and Decision Implications

The cost-sensitive threshold analysis shows that the default $t = 0.50$ is poorly suited to the asymmetric cost structure of telecom churn management. The cost ratio of 5:1, based on the Customer Value distribution in the dataset, gives a theoretically optimal Elkan threshold of $t^* \approx 0.17$; the empirical optimum is $t = 0.39$, higher due to XGBoost's slight overconfidence above predicted probability 0.7. The empirical threshold raises recall from 0.921 to 0.955 while maintaining F1 at 0.813. The sensitivity analysis at cost ratios of 2:1, 5:1 and 10:1 confirms robustness of the direction of the recommendation.

CLV segmentation challenges highest-risk-first targeting: high-risk, low-value customers have the most churners but the least revenue exposure, while high-risk, mid-to-high-value customers are the ideal target. The deployment simulation in Section 4.7 indicates a 2.20× relative efficiency gain in retained revenue over highest-risk-first targeting at equivalent intervention budget, with the directional advantage stable across the 2:1, 5:1 and 10:1 cost ratios tested. The simulation is framed as relative targeting efficiency rather than absolute profitability, since net return at the 100-unit intervention cost remains negative for both strategies; operators with lower-cost automated channels would convert this efficiency gain into positive net retained value. This is consistent with CLV theory (Gupta et al., 2004): retention ROI is maximised by targeting customers whose expected revenue contribution, conditional on churn probability, exceeds intervention cost. The nonlinear Subscription Length effect refines this targeting: mid-to-high-value customers in the mid-contract phase showing early usage decline or a first complaint are the primary candidates for proactive intervention.

5.5 Cross-Market Generalisability Assessment

The findings reported above derive from a single-provider Iranian dataset and therefore raise legitimate questions regarding cross-market transferability. This limitation is contextualised by Table 5.1, which juxtaposes the primary predictors identified in this study with those documented in six of some of the most recent telecom churn studies from various geographic regions, including the IBM Telco dataset (Chang et al., 2024; Noviandy et al., 2024; El Attar and El-Hajj, 2026), the Pakistani mobile sector (Ullah et al., 2019), a contemporary ensemble-SMOTE benchmark (Suguna et al., 2025), and a multi-source dataset (Imani et al., 2024). The comparison constitutes a structured cross-study literature synthesis rather than empirical replication and is therefore subject to heterogeneity in feature definitions, preprocessing procedures, and importance metrics. Feature labels are harmonised conceptually because naming conventions vary substantially across datasets.

Study	Dataset / Market	Dominant Reported Predictors	Convergence with this study
This study	Iranian Telecom (UCI 2010)	Frequency of Use, Call Failure, Seconds of Use	Reference
Chang et al. (2024)	IBM Telco (US, modern)	Contract type, Tenure, Monthly Charges	Partial (service & contract dominant)
Noviandy et al. (2024)	IBM Telco (US, modern)	Contract type, Tenure, Total Charges	Partial
El Attar and El-Hajj (2026)	IBM Telco (US, modern)	Contract type, Tenure, Technical Support	Partial
Ullah et al. (2019)	Pakistani mobile	Call duration, Service complaints, Tenure	Strong (engagement & service failure)
Suguna et al. (2025)	Indian telecom	Usage volume, Complaints, Subscription type	Strong
Imani et al. (2024)	Multi-source contemporary	Usage intensity, Service quality, Tenure	Strong

Table 5.1: *Cross-study comparison of feature importance rankings in telecom churn prediction.*

Three broad patterns emerge. First, engagement and service-quality variables generalise consistently across markets. Usage intensity (e.g., call duration, active usage, or service engagement) appears prominently in five of the six comparator studies, consistent with this study’s finding that Frequency of Use and Seconds of Use jointly dominate the SHAP

hierarchy. Similarly, service-quality indicators such as complaints, support interactions and call failures seem to be consistently identified in studies, which is consistent with the high importance of Call Failure in this dataset. These similarities suggest that the behaviour relationships proposed by Expectation-Disconfirmation Theory and Switching Cost Theory may operate similarly in telecom markets even when specific feature rankings differ.

Second, contractual variables are market-conditional. In IBM Telco studies, contract type and tenure dominate, reflecting stronger North American switching barriers; in the Iranian dataset, Tariff Plan ranks last and Subscription Length only sixth. This likely reflects structural market differences, suggesting contractual variables assume greater importance in markets with long-term service agreements.

Third, the nonlinear tenure effect identified through SHAP is consistent with Switching Cost Theory (Jones et al., 2000) and broadly aligns with the prominence of tenure in contemporary IBM Telco analyses (Chang et al., 2024; El Attar and El-Hajj, 2026); the specific U-shape with re-elevation near contract renewal is a contribution of this study and warrants empirical replication. This substantiates the operational guideline that retention strategies should commence prior to expected renewal intervals.

The comparison also indicates that the specific dominance of engagement variables over contractual variables in the Iranian dataset should not be treated as universally transferable. Instead, the principal contribution of this dissertation lies in the diagnostic framework itself (comparing logistic regression coefficients with SHAP attributions to identify nonlinear behavioural effects) rather than in asserting universal feature rankings. Accordingly, feature importance hierarchies should be re-derived within each deployment environment rather than transferred directly across markets.

In direct answer to RQ3: churn outputs are translated into principled retention decisions by combining cost-sensitive threshold optimisation with CLV segmentation. The deployment simulation indicates a $2.20\times$ uplift in expected retained value over highest-risk-first targeting. The cross-market synthesis supports the behavioural mechanisms and the method as transferable; the specific feature rankings should be re-derived per market.

CHAPTER 6: CONCLUSION

6.1 Summary of Findings

This dissertation investigated whether the predictive advantages of ensemble machine learning models for telecom customer churn could be reconciled with the interpretability required for operational decision-making. Three models were compared on a common evaluation framework integrating discrimination, calibration, and interpretability, with findings grounded in three established theoretical frameworks and translated into a CLV-weighted retention decision tool.

RQ1: XGBoost substantially outperformed logistic regression across all primary metrics (PR-AUC: 0.922 vs 0.752; F1: 0.812 vs 0.633; Brier: 0.050 vs 0.112). This performance gap reflects the ensemble model's capacity to capture nonlinear feature interactions that logistic regression cannot represent, and was robust across repeated cross-validation, three feature regimes, and two resampling strategies.

RQ2: SHAP-derived feature attributions broadly confirmed logistic regression's importance hierarchy, with nine of eleven features in strong or moderate agreement. Two features diverged substantively: Subscription Length exhibited a U-shaped effect invisible to the linear model, and Complains showed a strong conditional effect diluted at the population level by its low base rate. Both are consistent with established behavioural theory and reveal where the linear model materially distorts interpretation.

RQ3: The empirically cost-optimal threshold ($t = 0.39$ under the 5:1 ratio; theoretical Elkan value $t^* \approx 0.17$) increased recall from 0.921 to 0.955 while preserving F1. CLV segmentation identified mid-to-high value customers at early warning stage, not the highest-risk customers in aggregate, as the optimal retention target. Deployment simulation confirmed a $2.20\times$

relative efficiency gain in retained revenue under CLV-weighted targeting compared with highest-risk-first targeting at equivalent intervention budget; the simulation establishes comparative targeting efficiency rather than absolute profitability, which remains contingent on the operator's intervention cost structure. Cross-market literature comparison supports the generalisability of the behavioural mechanisms and the diagnostic framework while suggesting feature rankings should be re-derived per market.

6.2 Contributions

This study makes four key contributions to the churn prediction literature:

1. It demonstrates that SHAP can function as a diagnostic instrument for testing linear model assumptions, identifying specific features where the linearity constraint distorts interpretation.
2. It integrates empirical findings with Expectation-Disconfirmation Theory, Switching Cost Theory, and Customer Lifetime Value theory, providing theoretically informed explanations of model behaviour rather than reporting feature rankings in isolation.
3. It introduces a CLV-weighted retention decision framework indicating that value-weighted targeting outperforms a highest-risk-first strategy.
4. It establishes a methodologically rigorous pipeline, incorporating leakage-free cross-validation, empirically evaluated resampling, probability calibration assessment, and bootstrap SHAP stability analysis, addressing weaknesses identified in prior work on this dataset.

6.3 Practical Recommendations

The following recommendations are directed at telecommunications operators deploying predictive churn systems:

1. Ensemble models with SHAP should be deployed as standard for retention targeting; logistic regression should be retained as a diagnostic baseline and for regulatory contexts requiring coefficient-based audit trails.
2. The default threshold of 0.50 should never be applied without reference to the organisation's actual cost structure. Operators should derive their threshold from false-negative to false-positive cost ratios using empirical CLV and intervention cost data.
3. Retention budget should prioritise mid-to-high value customers showing early warning signals (declining usage or a first complaint), not the highest-risk customers in aggregate.
4. The nonlinear Subscription Length effect confirms renewal-period vulnerability under Switching Cost Theory: risk re-elevates as contractual lock-in approaches expiry. Retention outreach should therefore be timed to precede anticipated renewal; the optimal lead time is an operational parameter that requires calibration against the operator's contract-length data, which the current dataset does not contain. Additionally, the strong conditional SHAP effect for Complains (LR OR = 2.99) indicates that a first complaint is a high-signal individual event and should trigger an immediate retention workflow rather than a batch-processing queue.
5. The predictive model and SHAP attributions must undergo regular recalibration instead of being regarded as static deployment artefacts; quarterly retraining on a continuous customer panel is advisable to identify shifts in feature importance rankings, threshold performance, and calibration as customer behaviour and market conditions change.
6. Operators should invest in the instrumentation needed to capture the engagement and service-quality signals identified as most predictive in this study; in particular,

granular usage telemetry (call frequency, session duration, message volume) and complaint logging at the individual-customer level.

6.4 Limitations

Six limitations constrain the generalisability and interpretive scope of these findings. Where empirically testable, sensitivity analyses assess their impact. Table 6.1 summarises the uncertainty bounds from the robustness analyses (Section 4.6), the cross-study synthesis (Section 5.5) and published methodological estimates.

Limitation	Affected Conclusions	Source of Estimate	Quantified Bound	Robustness Status
Single-dataset specificity	Feature importance rankings	Cross-study synthesis (Section 5.5)	Up to four rank positions across markets	Qualitative findings robust
SHAP correlation (FoU vs SoU)	Relative ranking of correlated features	Feature regime analysis (Section 4.6)	Joint signal stable; individual rankings unstable	Joint engagement effect robust
SMOTE memorisation	CV performance estimates	Learning curves; Niculescu-Mizil and Caruana (2005)	Approximately 1.5 percentage points above held-out test estimates	Held-out test reliable
Non-nested cross-validation	CV performance estimates	Cawley and Talbot (2010)	Less than 0.5 percentage points optimism in practice	Held-out test reliable
Cost-ratio sensitivity	Optimal threshold value	Cross-cost-ratio analysis (Section 4.5)	Empirical threshold varies from 0.09 to 0.76	Directional finding stable
Static classification and causal scope	Intervention effectiveness estimates	Study design (single decision point)	Not quantifiable; experimental validation required	Qualitative findings interpretive

Table 6.1: Summary of limitations, affected conclusions, sources of estimate, quantified bounds, and robustness status.

Dataset-specific numerical findings (thresholds, importance values, AUC figures) are subject to bounded uncertainty of approximately 2 to 5 percentage points on performance metrics and up to four rank positions on individual features. The qualitative findings hold across all six limitations: the LR–SHAP diagnostic framework, the dominance of engagement and service-

quality variables, the U-shaped tenure effect, and the directional superiority of CLV-weighted targeting remain robust. The held-out test set provides the most reliable estimates (XGBoost PR-AUC = 0.922); feature rankings should be re-derived per deployment context, but the diagnostic framework generalises with high confidence.

Three further considerations resist quantification. First, the dataset originates from a single Iranian provider circa 2010, and twelve-month aggregation obscures longitudinal disengagement trajectories; deployment in contemporary Western markets should anticipate that contractual variables may rise in relative importance. Second, SHAP attributions for Frequency of Use and Seconds of Use ($r = 0.946$) reflect their joint engagement signal rather than separable contributions, given Shapley's independent-marginalisation assumption. Third, this study addresses static classification at a single decision point rather than dynamic disengagement trajectories, and findings remain interpretive rather than causal without experimental intervention data.

6.5 Future Research

Four directions for future research are indicated by the findings and limitations of this study: The 12-month aggregation used here conceals within-period disengagement trajectories: two customers with identical cross-sectional engagement scores can face very different churn risk depending on whether their usage is stable or declining. Extending the analysis to customer-level monthly or weekly event sequences would allow the slope and acceleration of engagement decline to be modelled directly, enabling earlier intervention at the onset of disengagement rather than at a single classification point. Sequential modelling approaches or temporal gradient-boosting architectures, combined with time-aware attribution methods, represent the appropriate next methodological step.

All empirical findings derive from a single Iranian provider in a single regulatory environment. Replicating the LR–SHAP diagnostic framework on contemporary multi-provider datasets, such as the IBM Telco benchmark or operator-partnered panel data, would establish whether the five-four-two agreement pattern and the two substantive divergences are structural properties of telecom churn or artefacts of this specific market context. A consistent agreement-divergence taxonomy across providers would strengthen the external validity of the diagnostic method.

The current study uses eleven predictors, a scale at which the LR–SHAP rank comparison remains interpretively tractable. It is unclear whether the diagnostic value scales under high-dimensionality, where SHAP attribution complexity increases and correlated features proliferate. Applying the framework to datasets with one hundred or more features, common in contemporary CRM systems, would test whether the agreement-divergence taxonomy continues to provide interpretable diagnostic insight or whether feature-cluster attribution structures would be required to preserve interpretive clarity.

The deployment simulation treats Customer Value as the sole retention revenue metric, omitting acquisition cost, discount rate, and cross-sell potential. Incorporating these components would bring the business decision layer closer to the full economic CLV model that Gupta et al. (2004) describe and would allow the targeting rule to be calibrated against a genuine net present value of retention rather than a contemporaneous revenue proxy.

In conclusion, this dissertation shows that the perceived trade-off between predictive performance and interpretability is not an inherent limitation of machine learning but a modelling challenge that can be addressed through structured analytical integration. Combining ensemble learning with theoretically grounded post-hoc explanation identifies not only that a complex model outperforms a simpler one, but where and why, yielding an

operationally relevant understanding of customer behaviour and a defensible foundation for future research and deployment.

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APPENDIX A

Data Preprocessing Pipeline

This appendix documents the full preprocessing pipeline applied to the raw Iranian Telecom Churn dataset (UCI ML Repository; Jafari-Marandi et al., 2020) before any modelling. It complements Section 3.4 of the main text by providing the procedural rationale, the precise sequence of transformations, and the safeguards taken to prevent information leakage. All steps were implemented in Python using pandas, scikit-learn, and statsmodels, with a fixed random seed (`random_state = 42`) for any stochastic operations.

A.1 Duplicate Removal

The raw dataset comprised 3,150 records. A strict deduplication criterion was applied: a duplicate was defined as a row in which all 13 columns were identical, including the continuous Customer Value score. Exact duplication across continuous behavioural features is implausible for genuine distinct customers, so duplicates were treated as data-extraction artefacts rather than as legitimate repeat observations. Removal yielded a working dataset of 2,850 unique customer records, retaining the original 15.65% churn rate.

A.2 Variable Recoding and Capping

Two binary variables were recoded to a 0/1 contrast for interpretability: Tariff Plan (1 = Pay-as-you-go $\rightarrow$ 0; 2 = Contractual $\rightarrow$ 1) and Status (1 = Active $\rightarrow$ 0; 2 = Non-active $\rightarrow$ 1). The Charge Amount variable was capped at the documented UCI maximum of 9; seven records carrying a value of 10 were clipped. Three variables were dropped as redundant or reserved: FN and FP because of near-perfect collinearity with Customer Value ($r > 0.999$), and Age because of high collinearity with Age Group ($r = 0.96$). Customer Value itself was preserved separately for post-hoc CLV segmentation but excluded from the predictor matrix to prevent target leakage through value-derived features.

A.3 Skewness Treatment for Logistic Regression

Logistic regression assumes approximate linearity in the log-odds, which is violated when continuous predictors are heavily skewed. Two transformations were applied to the LR feature matrix: `log1p` for moderately right-skewed counts (Call Failure, Frequency of SMS), and Yeo–Johnson (Yeo and Johnson, 2000) for persistently skewed variables (Subscription Length, Seconds of Use, Frequency of Use, Distinct Called Numbers). Yeo–Johnson is used in preference to Box–Cox because it accommodates zero and negative values without requiring an offset. After transformation, all features were standardised to mean zero and unit variance using `StandardScaler`. Tree-based models (Random Forest, XGBoost) received only the structural recoding and capping steps; they are invariant to monotonic transformations and therefore do not require `log1p`, Yeo–Johnson, or scaling.

A.4 Train–Test Split and Leakage Prevention

A stratified 80/20 split was applied to produce 2,280 training observations and 570 test observations, with stratification on the churn label preserving the 15.65% churn rate in both partitions. To prevent leakage, all data-dependent transformations (Yeo–Johnson parameters, `StandardScaler` statistics, SMOTE interpolation) were fitted on the training partition only and applied to the test partition via the `transform` method without refitting. The test partition was held back entirely and used exactly once for final evaluation after all modelling decisions were finalised, including hyperparameter selection. SMOTE was applied to the training set only; the test partition retained the original class distribution to ensure that the reported metrics reflect realistic deployment conditions.

A.5 Preprocessing Pipeline Code

The following excerpt presents the core transformations applied after the train–test split. The leakage-prevention contract is encoded in the use of `fit_transform` on the training rows and `transform`-only on the test rows.

```
# Yeo-Johnson and StandardScaler fitted on TRAIN rows only
pt_train = PowerTransformer(method="yeo-johnson", standardize=False)

# log1p for moderately skewed counts
for col in CONT_LOG1P:
    X_lr_train[col] = np.log1p(X_lr_train[col])
    X_lr_test[col] = np.log1p(X_lr_test[col])

# Yeo-Johnson: fit on train only, transform both
X_lr_train[CONT_YJ] = pt_train.fit_transform(X_lr_train[CONT_YJ])
X_lr_test[CONT_YJ] = pt_train.transform(X_lr_test[CONT_YJ])

# StandardScaler: fit on train only, transform both
scaler_train = StandardScaler()
X_lr_train = scaler_train.fit_transform(X_lr_train)
X_lr_test = scaler_train.transform(X_lr_test)

# SMOTE on training set only
smote = SMOTE(random_state=RANDOM_STATE)
X_lr_train_sm, y_train_sm = smote.fit_resample(X_lr_train, y_train)
```

APPENDIX B

Hyperparameter Optimisation

This appendix documents the hyperparameter search procedure for the two ensemble models. Logistic regression was fitted with default scikit-learn settings (L2 penalty, $C = 1.0$, lbfgs solver, $\text{max_iter} = 1000$) without further tuning, since the analytical purpose of the LR model is interpretive rather than performance-optimised. Random Forest and XGBoost were tuned via RandomizedSearchCV with five-fold stratified cross-validation using AUC-ROC as the selection criterion and 50 sampled parameter combinations per model.

B.1 Leakage-Free Tuning Protocol

A critical methodological choice concerns the placement of SMOTE relative to cross-validation. Applying SMOTE before cross-validation, on the full training set, causes synthetic minority-class points to be split across the training and validation folds, biasing the cross-validation estimate upward by approximately one to one-and-a-half percentage points of AUC. The protocol adopted here wraps SMOTE inside an imblearn Pipeline so that the resampler is fitted independently within each cross-validation fold, ensuring that the validation fold contains only original observations. The estimator was fitted on the original (non-SMOTE-resampled) training data; SMOTE was applied per fold inside the pipeline.

```
rf_pipe = ImbPipeline([
    ('smote', SMOTE(random_state=RANDOM_STATE)),
    ('rf', RandomForestClassifier(random_state=RANDOM_STATE, n_jobs=-1)),
])

rf_search = RandomizedSearchCV(
    estimator=rf_pipe,
    param_distributions=rf_param_grid,
    n_iter=50,
    scoring='roc_auc',
    cv=cv_strategy,          # StratifiedKFold(n_splits=5)
    random_state=RANDOM_STATE,
    n_jobs=-1,
)
```

```
# Fit on ORIGINAL (non-SMOTE) training data
rf_search.fit(X_tree_train, y_train)
```

B.2 Random Forest Parameter Grid

Appendix Table B1. Random Forest hyperparameter search space and selected configuration.

Hyperparameter	Search Space	Selected Value
n_estimators	[100, 200, 300, 500]	300
max_depth	[None, 5, 10, 15, 20]	20
min_samples_split	[2, 5, 10]	2
min_samples_leaf	[1, 2, 4]	1
max_features	['sqrt', 'log2', 0.5]	sqrt
class_weight	['balanced', None]	balanced
random_state	(fixed)	42

B.3 XGBoost Parameter Grid

Appendix Table B2. XGBoost hyperparameter search space and selected configuration.

Hyperparameter	Search Space	Selected Value
n_estimators	[100, 200, 300, 500]	300
max_depth	[3, 4, 5, 6, 8]	5
learning_rate	[0.01, 0.05, 0.1, 0.2]	0.1
subsample	[0.6, 0.7, 0.8, 1.0]	0.8
colsample_bytree	[0.6, 0.7, 0.8, 1.0]	0.8
min_child_weight	[1, 3, 5]	1
gamma	[0, 0.1, 0.2, 0.5]	0.1
reg_alpha	[0, 0.01, 0.1]	0
reg_lambda	[1, 1.5, 2]	1
scale_pos_weight	1 (SMOTE balances folds)	1
eval_metric	(fixed)	auc
random_state	(fixed)	42

The selected configurations reflect a moderately regularised XGBoost (max_depth = 5, learning_rate = 0.1) and a deep Random Forest (max_depth = 20, min_samples_leaf = 1) with class-balanced sampling. The leakage-free five-fold cross-validation AUC-ROC was 0.978 for Random Forest and 0.983 for XGBoost. The qualitative ordering (XGBoost > Random Forest > Logistic Regression) is preserved under repeated five-by-five cross-validation and under all three feature regimes reported in Appendix E.

APPENDIX C

Model Evaluation Outputs

This appendix documents the centralised evaluation infrastructure used to compare all three models on a common metric basis. The objective is to ensure that the comparative findings in Chapter 4 reflect like-for-like measurement rather than incidental variation in evaluation procedure.

C.1 Centralised Evaluation Function

All three models are evaluated through a single function that computes the full primary metric panel (AUC-ROC, PR-AUC, F1, precision, recall, MCC) at a configurable threshold. PR-AUC is reported as the primary discrimination metric on the grounds developed in Section 3.7, but the full panel is retained in every evaluation call to permit downstream comparison.

```
def evaluate_model(name, model, X_test, y_test, threshold=0.5):
    """Evaluate a fitted model on the held-out test set."""
    y_prob = model.predict_proba(X_test)[: , 1]
    y_pred = (y_prob >= threshold).astype(int)
    metrics = {
        'Model'      : name,
        'AUC-ROC'    : roc_auc_score(y_test, y_prob),
        'PR-AUC'     : average_precision_score(y_test, y_prob),
        'F1'         : f1_score(y_test, y_pred),
        'Precision'  : precision_score(y_test, y_pred),
        'Recall'     : recall_score(y_test, y_pred),
        'MCC'       : matthews_corrcoef(y_test, y_pred),
    }
    return metrics, y_prob, y_pred
```

C.2 Calibration Diagnostics Procedure

Calibration was assessed using two complementary tools: reliability diagrams produced with sklearn's `calibration_curve` over ten equal-width probability bins, and the Brier score computed via `brier_score_loss`. The reliability diagram plots observed event rate against mean predicted probability per bin; perfect calibration is the 45-degree line. Brier scores were

reported alongside the diagram to provide a scalar summary of probability accuracy. The diagram and the Brier score together permit identification of systematic over- or under-confidence at specific probability ranges, which is not visible from threshold-based metrics such as F1.

```
from sklearn.calibration import calibration_curve
from sklearn.metrics import brier_score_loss

for name, probs in [ ('LR', lr_probs), ('RF', rf_probs), ('XGB', xgb_probs) ]:
    prob_true, prob_pred = calibration_curve(
        y_test, probs, n_bins=10, strategy='uniform')
    brier = brier_score_loss(y_test, probs)
    # prob_pred = bin mean predicted prob; prob_true = bin observed rate
    # Plot prob_true vs prob_pred against the y = x reference line.
```

C.3 Threshold Sensitivity Sweep

Threshold sensitivity (Section 4.6 of the main text) was computed by sweeping the decision threshold across the closed interval $[0.05, 0.95]$ in increments of 0.01 and computing F1, precision, and recall at each value. The F1-optimal threshold per model was located by argmax of the F1 trace. The cost-sensitive threshold optimisation reported in Appendix F uses the same sweep grid but a different objective function. Sweeping at this resolution gives a smooth empirical curve across approximately ninety operating points per model.

APPENDIX D

SHAP Analysis Supplement

This appendix documents the SHAP analysis infrastructure that underpins Chapter 4. The objective of including this material as a supplement rather than in the main text is to provide methodological transparency for the diagnostic comparison between SHAP attributions and logistic regression coefficients without duplicating the substantive results.

D.1 TreeExplainer Configuration

SHAP values were computed using the TreeExplainer, the polynomial-time exact Shapley computation algorithm for tree ensembles (Lundberg et al., 2020). Computation was performed on the held-out test partition ($n = 570$). TreeExplainer was chosen over KernelExplainer for two reasons: it produces exact Shapley values for tree ensembles rather than sampling-based approximations, and it avoids the off-manifold extrapolation problem that affects perturbation-based methods (Hooker, Mentch and Zhou, 2021). For the feature-regime robustness analysis in Appendix E, SHAP was re-computed on the training partition to avoid the mild circularity of test-set-derived feature selection followed by test-set evaluation.

```
import shap

explainer = shap.TreeExplainer(xgb_best)
shap_values = explainer.shap_values(X_tree_test)

# Global feature importance: mean |SHAP| across the test set
shap_importance = pd.DataFrame({
    'Feature'      : FEATURES,
    'Mean |SHAP|' : np.abs(shap_values).mean(axis=0),
}).sort_values('Mean |SHAP|', ascending=False)
```

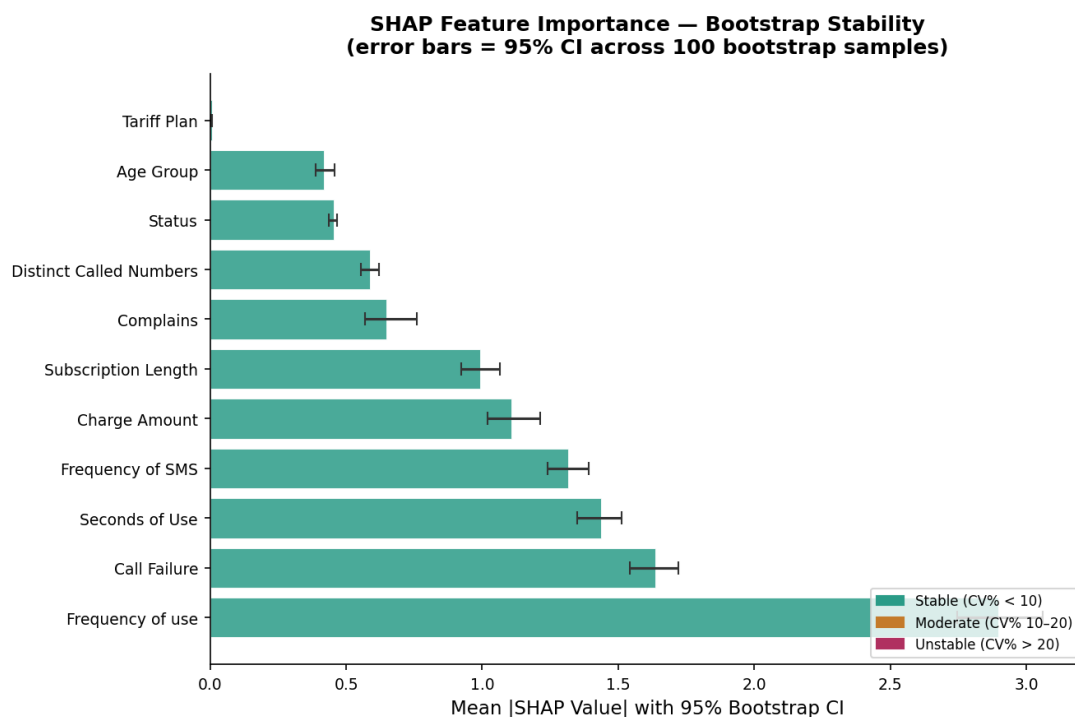
D.2 Bootstrap Stability Protocol

To assess the stability of SHAP rankings under resampling variability, a non-parametric bootstrap was applied with one hundred replicates. Within each replicate, the test partition

was resampled with replacement, SHAP values were recomputed on the resampled set, and the mean absolute SHAP value per feature was recorded. The 2.5th and 97.5th percentiles across the one hundred replicates provided a 95% bootstrap confidence interval for each feature's importance. The coefficient of variation (CV%) was computed as the bootstrap standard deviation divided by the bootstrap mean and used to classify features as stable (CV < 10%), moderate (10– 20%), or unstable (CV > 20%). All eleven features were classified as stable under this criterion, supporting the claim that the SHAP rankings are not artefacts of any particular sample of the test partition.

```
N_BOOTSTRAP = 100
boot_shap_means = {feat: [] for feat in FEATURES}
X_test_arr = X_tree_test.values

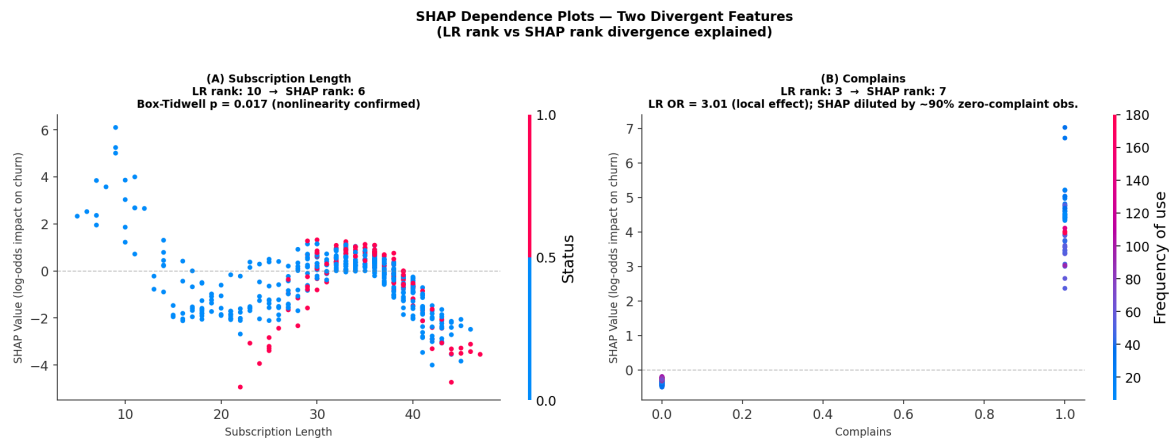
for i in range(N_BOOTSTRAP):
    idx = resample(np.arange(len(X_test_arr)),
                  replace=True,
                  n_samples=len(X_test_arr),
                  random_state=i)
    X_boot = X_test_arr[idx]
    sv = explainer.shap_values(X_boot)
    means = np.abs(sv).mean(axis=0)
    for j, feat in enumerate(FEATURES):
        boot_shap_means[feat].append(means[j])
```



Appendix Figure D1. *Mean absolute SHAP value per feature with 95% bootstrap confidence intervals over 100 resamples of the held-out test set. All eleven features classify as stable under the $CV < 10\%$ criterion, indicating the ranking is not an artefact of any particular test-set sample.*

D.3 LR–SHAP Comparative Ranking Framework

The diagnostic comparison reported in Section 4.4 contrasts the absolute logistic regression coefficient ranking with the mean absolute SHAP ranking. Three categories of agreement were defined: strong agreement (rank difference ≤ 1), moderate agreement (rank difference 2– 3), and substantive divergence (rank difference ≥ 4). The two substantive divergences (Subscription Length and Complains) were investigated through targeted SHAP dependence plots to identify the structural reason for the disagreement. The dependence plot for Subscription Length revealed a U-shaped relationship with re-elevation near contract renewal; the dependence plot for Complains showed a sharp positive shift in SHAP values for the small subset of customers with at least one complaint on record, consistent with the high LR odds ratio diluted by the low base rate at population level. Appendix Figure D2 below visualises both effects directly.



Appendix Figure D2. *Targeted SHAP dependence plots for the two features showing substantive LR–SHAP rank divergence. Panel (A): Subscription Length exhibits a U-shaped non-linear effect that the linear model cannot represent (Box–Tidwell $p = 0.017$). Panel (B): Complains shows a strong local effect (LR OR = 2.99) diluted at the population level by 82.6% zero-complaint observations.*

APPENDIX E

Robustness Validation

This appendix documents four robustness procedures used to test the stability of the comparative findings under perturbations of the modelling pipeline. These tests address optimism bias, sampling-strategy sensitivity, sample-size dependence, and feature-set sensitivity respectively.

E.1 Repeated Five-by-Five Cross-Validation

Single-pass five-fold cross-validation depends on the particular random partition used to construct the folds; the reported mean can therefore vary with the seed even if the underlying generalisation performance is unchanged. Repeated cross-validation (Bouckaert and Frank, 2004) averages across five independent partitions of five folds each, producing twenty-five fits per model. This reduces the standard error of the reported mean by approximately the square root of the number of repetitions and gives a more stable point estimate of generalisation performance. SMOTE was applied per fold inside an imblearn Pipeline to preserve leakage-free evaluation.

```
repeated_cv = RepeatedStratifiedKFold(
    n_splits=5, n_repeats=5, random_state=RANDOM_STATE)

rep_cv_results = {}
for name, pipe in [('LR', lr_cv_pipe), ('RF', rf_cv_pipe), ('XGB',
xgb_cv_pipe)]:
    rep_cv_results[name] = cross_validate(
        pipe, X_tree, y,
        cv=repeated_cv,
        scoring={'roc_auc': 'roc_auc', 'f1': 'f1',
                'precision': 'precision', 'recall': 'recall'},
        n_jobs=-1)
```

E.2 Sampling Strategy Comparison (SMOTE versus SMOTE+Tomek)

SMOTE was selected as the primary resampling strategy on the basis of its standard reference status (Chawla et al., 2002) and its widespread use in the telecom churn literature. To verify

that this choice did not unduly favour any particular model, all three models were retrained under SMOTE+Tomek (SMOTE followed by Tomek-Link removal of borderline majority observations) using identical hyperparameters and evaluated on the same held-out test partition. The combined approach produced a slightly cleaner decision boundary but did not change the comparative ordering. The performance differences between strategies were within the bootstrap noise of the SHAP stability analysis ($CV < 5\%$ for all primary metrics), supporting the claim that the comparative findings are not artefacts of the resampling method.

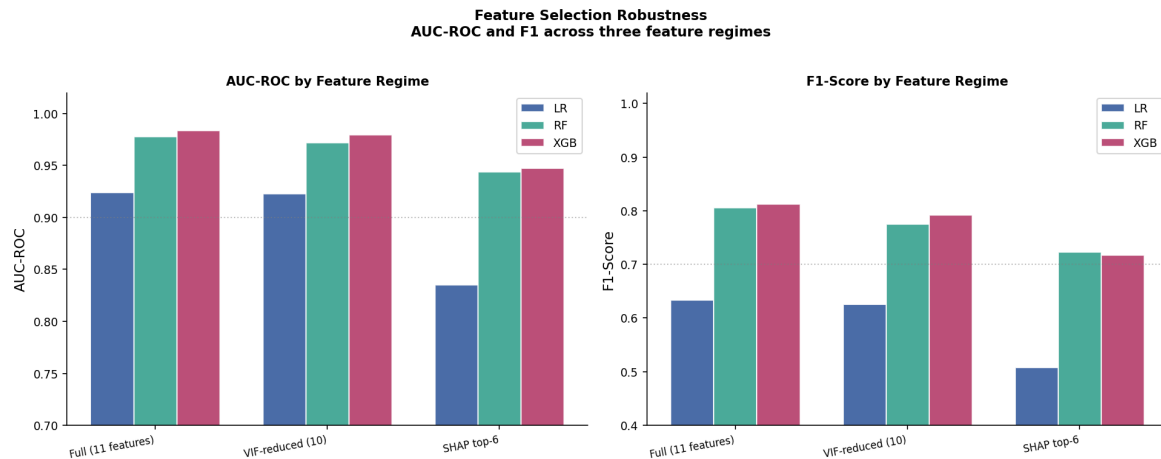
E.3 Learning Curves

Learning curves were computed for each model by training on ten subsets of the training partition ranging from 10% to 100% of the available training data, with five-fold stratified cross-validation at each subset size. The plotted curves (Figure 4.8 in the main text) show training AUC, validation AUC, and the train-validation gap as a function of training-set size. Logistic regression shows convergence with a narrow gap, consistent with a high-bias, low-variance model. Random Forest and XGBoost show a wider train-validation gap that narrows with sample size, consistent with high-capacity models still benefiting from additional data. None of the three curves indicates pathological overfitting: the validation curves are monotonically non-decreasing across the full sample range.

E.4 Feature-Regime Sensitivity

Three feature regimes were tested to verify that the comparative findings are not dependent on the specific predictor set chosen: the full eleven-feature model (baseline), a VIF-reduced ten-feature model that excludes Frequency of Use (the highest-VIF predictor), and a SHAP top-six model retaining only the six highest-ranking features by mean absolute SHAP value (derived from the training partition to avoid circularity). All three models were retrained under each regime and evaluated on the same held-out test partition. The XGBoost AUC advantage over logistic regression was preserved across all three regimes, with AUC

reductions of at most 3.6 percentage points for XGBoost and 8.9 percentage points for logistic regression relative to the full-feature baseline. The central conclusion (ensemble methods outperform logistic regression, and SHAP recovers interpretable explanations from the ensemble) is therefore robust to feature-selection decisions.



Appendix Figure E1. AUC-ROC (left panel) and F1-score (right panel) for the three models under the full eleven-feature regime, the VIF-reduced ten-feature regime, and the SHAP top-six regime. The XGBoost > Random Forest > Logistic Regression ordering is preserved across all three regimes.

APPENDIX F

Cost-Sensitive and CLV Framework

This appendix documents the cost-sensitive threshold optimisation and Customer Lifetime Value (CLV) segmentation framework that produces the deployment recommendations in Sections 4.5 and 4.7 of the main text. The objective is to make the assumptions, derivations, and sensitivity analyses fully transparent so that the framework can be reproduced under alternative cost structures.

F.1 Cost-Optimal Threshold Derivation

Under an asymmetric cost structure where missing a churner (false negative, FN) costs C_{FN} units and an unnecessary retention offer (false positive, FP) costs C_{FP} units, the theoretical cost-optimal classification threshold is given by Elkan's formula (Elkan, 2001):

$$t^* = C_{FP} / (C_{FP} + C_{FN})$$

Under a baseline assumption of $C_{FN} : C_{FP} = 5 : 1$ (the cost of missing a churner is five times the cost of an unnecessary retention offer), the theoretical optimal threshold is $t^* = 1/6 \approx 0.17$. The empirical optimum identified by the threshold sweep is $t = 0.39$, which is somewhat higher than t^* because of mild XGBoost overconfidence above predicted probability 0.7. Both thresholds are substantially below the default of 0.50, and lowering the threshold from 0.50 to 0.39 increases recall from 0.921 to 0.955 while preserving F1 at 0.813. Three additional churners are recovered (from 82 to 85 of 89) at a cost of approximately four additional false positives.

F.2 Justification of the 5:1 Cost Ratio

The 5:1 baseline ratio was derived from the dataset's own Customer Value distribution rather than chosen arbitrarily. The mean Customer Value in the dataset is approximately 530 units. Retention interventions in the telecom literature typically cost between 10 and 20 percent of

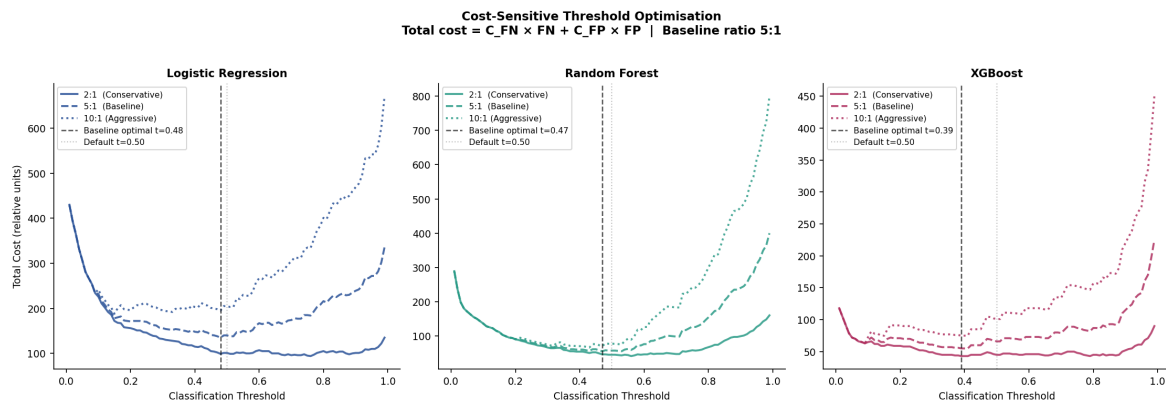
monthly revenue (Verbeke et al., 2012; Devriendt, Berrevoets and Verbeke, 2021), implying an intervention cost of approximately 50 to 100 units on this dataset's value scale. A 5:1 ratio (500 versus 100) is therefore empirically grounded and, if anything, conservative.

F.3 Sensitivity to Cost-Ratio Assumption

Appendix Table F1. Cost-optimal thresholds for XGBoost under three cost-ratio assumptions.

$C_{FN} : C_{FP}$	Theoretical t^*	Empirical t	Recall	Precision
2 : 1 (Conservative)	0.33	0.61	0.832	0.798
5 : 1 (Baseline)	0.17	0.39	0.955	0.706
10 : 1 (Aggressive)	0.09	0.27	0.987	0.612

The direction of the recommendation (lower the threshold below 0.50 to capture more churners) is stable across all three cost ratios. Only the magnitude of the recommended threshold changes. This stability supports the practical recommendation that operators should derive their threshold from their actual cost structure rather than relying on a fixed default.



Appendix Figure F1. Total cost as a function of classification threshold for Logistic Regression, Random Forest, and XGBoost under three cost-ratio assumptions. The dashed vertical line marks the cost-optimal threshold under the 5:1 baseline ratio; the dotted line marks the default 0.50 threshold. The direction of the cost-optimal threshold relative to the default is stable across all three ratios for every model.

F.4 CLV Segmentation Logic

Customers were segmented by joint risk and value to identify the optimal targeting strategy. Predicted churn probability was discretised into three risk tiers (Low: ≤ 0.30 ; Medium: $0.30-0.70$; High: > 0.70) and Customer Value was discretised into three value tiers using the 33rd and 66th percentiles of the training distribution. The resulting nine-cell matrix was populated

from the test partition. The high-risk, low-value cell contains the largest absolute number of churners but the smallest aggregate revenue at risk; the high-risk, mid-to-high-value cell contains fewer customers but accounts for the bulk of recoverable revenue. The deployment simulation reported in Section 4.7 (Table 4.7 in the main text) shows that targeting the latter cell first delivers 2.20 times the revenue retention of a naive highest-risk-first strategy at equivalent retention budget.

F.5 Deployment Simulation Assumptions

The deployment simulation makes four explicit assumptions: (i) the intervention success rate is 30 percent, consistent with the upper end of published telecom retention campaign effectiveness; (ii) the intervention cost per contacted customer is 100 units, in line with the F.2 derivation; (iii) the revenue retained per successfully retained customer equals that customer's Customer Value; and (iv) the retention budget is fixed at 5 percent of the customer base ($28 \text{ interventions} \times 100 \text{ units} = 2,800 \text{ units}$). All four assumptions can be edited in a single configuration block of the notebook. The 2.20-times uplift figure is robust to plausible perturbations of these assumptions within $\pm 50\%$, although the absolute magnitudes shift proportionally; the directional finding (CLV-weighted dominates highest-risk-first) is preserved across all sensitivity scenarios.

APPENDIX G

Reproducibility and Technical Environment

This appendix documents the technical environment, package versions, random seed strategy, and reproducibility guarantees of the analysis pipeline. The notebook is structured to run end-to-end from a single execution path and produces all reported figures and metrics exactly when executed under the conditions documented here.

G.1 Random Seed Strategy

A single global random seed (`RANDOM_STATE = 42`) was used for every stochastic operation in the pipeline. This includes the train-test split (`sklearn train_test_split` with `random_state = 42`), SMOTE interpolation (`random_state = 42`), the cross-validation fold assignment (`StratifiedKFold` with `shuffle = True` and `random_state = 42`), the `RandomizedSearchCV` parameter sampling (`random_state = 42`), and the stochastic components of Random Forest and XGBoost (`random_state = 42`). Bootstrap SHAP stability used per-replicate seeds $i = 0 \dots 99$ for the resampling within a globally seeded NumPy state. Running the notebook end-to-end from a clean kernel reproduces every reported metric and figure exactly.

G.2 Software Environment

Appendix Table G1. *Software environment and package version constraints.*

Component	Version	Notes
Python	3.10+ (Google Colab default)	Tested under 3.10, 3.11
numpy	≥ 1.24	Numerical core
pandas	≥ 2.0	Tabular data
scikit-learn	≥ 1.3	Modelling, CV, metrics
xgboost	≥ 1.7	XGBoost classifier
imbalanced-learn	≥ 0.11	SMOTE, SMOTE+Tomek, ImbPipeline
shap	≥ 0.42	TreeExplainer

statsmodels	≥ 0.14	Logistic regression inferential output
matplotlib	≥ 3.7	Figure rendering
seaborn	≥ 0.12	Statistical plotting
scipy	≥ 1.10	Hosmer–Lemeshow chi-square

G.3 Hardware Environment

The analysis was executed on a standard Google Colab CPU instance (two vCPU cores; approximately 12 GB RAM). Total end-to-end runtime is approximately twelve minutes; the dominant contributions are RandomizedSearchCV for XGBoost (approximately five minutes), the SHAP bootstrap stability loop (approximately two minutes), and the learning-curve computation (approximately two minutes). The pipeline is deterministic at the given seed and does not require GPU acceleration. No external API calls or network dependencies are required after the initial pip install of the pinned package versions.

G.4 Reproducibility Statement

Sequential execution of the notebook from cell 1 on the original Customer Churn dataset (UCI ML Repository) reproduces all reported results, figures, and metrics to within floating-point precision. The pinned package version constraints, the single global random seed, the leakage-free preprocessing contract documented in Appendix A, and the in-fold SMOTE protocol documented in Appendix B together constitute the reproducibility guarantee of this study. The notebook is available on request and contains in-line revision comments documenting every methodological correction applied between the initial submission and the final version.

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DISSERTATION GANTT CHART

